

**CENTER FOR DRUG EVALUATION AND
RESEARCH**

APPLICATION NUMBER:

201110Orig1s000

**CLINICAL PHARMACOLOGY AND
BIOPHARMACEUTICS REVIEW(S)**

Office of Clinical Pharmacology Review

NDA or BLA Number	NDA 201110
Link to EDR	\\CDSESUB1\evsprod\NDA201110\201110.enx
Submission Date	10/29/2019; 4/7/2020
Submission Type	Resubmission/Class 2
Brand Name	MILPROSA™
Generic Name	Progesterone
Dosage Form and Strength	Vaginal system containing 1.78 grams of progesterone and releases an average of 11 mg/day progesterone over 7 days
Route of Administration	Inserted vaginally
Proposed Indication	Support embryo implantation and early pregnancy (up to 10 weeks post-embryo transfer) by supplementation of corpus luteal function as part of an Assisted Reproductive Technology (ART) treatment program for infertile women
Applicant	Ferring Pharmaceuticals Inc.
Associated IND	IND 070875
OCP Review Team	Peng Zou, PhD; Yanhui Lu, PhD
OCP Final Signatory	Yanhui Lu, PhD Team leader Division of Cardiometabolic and Endocrine Pharmacology (DCEP) Office of Clinical Pharmacology

Memorandum:

The original application and resubmission of NDA 201110 submitted on April 30, 2010 and February 25, 2016, respectively, were reviewed by the Office of Clinical Pharmacology/Division of Clinical Pharmacology 3 and found acceptable from a Clinical Pharmacology perspective. For details, refer to the Clinical Pharmacology reviews by Dr. Sandhya Apparaju dated 02/02/2011 and by Dr. Peng Zou dated 09/07/2016 in DARRTS. No new approvability issues were identified in this resubmission. Therefore, the clinical pharmacology review team in DCEP recommends approval from the clinical pharmacology standpoint.

In this resubmission, the Applicant provided literature information to support the labeling language for Distribution, Metabolism, and Excretion under Section 12.3 Pharmacokinetics and changed the regulatory pathway from 505(b)(1) to 505(b)(2). A separate labeling review will be uploaded to DARRTS regarding our detailed labeling recommendations.

This is a representation of an electronic record that was signed electronically. Following this are manifestations of any and all electronic signatures for this electronic record.

/s/

PENG ZOU
04/08/2020 04:26:34 PM

YANHUI LU
04/08/2020 04:29:32 PM

OFFICE OF CLINICAL PHARMACOLOGY REVIEW

NDA: 201110	Submission Dates: 04/30/2010, 08/13/2010, 02/25/2016
Proposed Brand Name:	Milprosa™
Generic Name:	Progesterone
Clinical Pharmacology Primary Reviewer:	Peng Zou, PhD
Clinical Pharmacology Secondary Reviewer:	Doanh Tran, PhD
OCP Divisions:	Division of Clinical Pharmacology 3 (DCP3)
OND Division:	Division of Bone, Reproductive, and Urologic Products (DBRUP)
Sponsor:	Ferring Pharmaceuticals Inc.
Submission Type:	Original NDA/Resubmission
Relevant IND	IND 70875
Formulation, Strength, and Dosing Regimen	Vaginal ring for once-weekly use; (b) (4)
Indication:	Progesterone supplementation for up to 10 weeks post-embryo transfer as part of Assisted Reproductive Technology (ART) treatment in infertile women

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1 EXECUTIVE SUMMARY

The original sponsor, Teva Women's Health Inc. ("Teva"), submitted a 505(b)(1) New Drug Application (NDA) for Progesterone Vaginal Ring on April 30, 2010. The ownership of NDA 201110 was transferred from Teva to Ferring Pharmaceuticals Inc. ("Ferring") on August 5, 2015. Each weekly vaginal ring (DR-201, formerly DR-2011) consists of ~1.8 g of progesterone in a silicone matrix at a concentration of (b) (4). It is designed to deliver a prolonged and controlled steady release of progesterone and has a progesterone release rate of approximately 11 mg/day for up to 7 days, based on evaluations of residual progesterone in the vaginal ring after patient use in a clinical study.

In the submissions dated 04/30/2010 and 08/13/2010, Teva conducted 7 phase 1, 2 & 3 clinical trials in order to assess progesterone pharmacokinetics (PK; systemic and tissue progesterone), pharmacodynamics (PD; successful secretory transformation of the endometrium) and efficacy (clinical pregnancy rates) with the progesterone vaginal ring formulation. In addition, bioequivalence (BE) of the clinical vs. commercial formulations (change in site of manufacturing only) and estimation of in vivo drug release have been addressed. The BE Study 10936035 conducted in healthy post-menopausal females demonstrated the BE between the clinical trial formulation and the commercial formulation. The clinical pharmacology data was reviewed by Dr. Sandhya Apparaju and found acceptable on 02/02/2011.

In the Complete Response letter issued on 2/28/2011, DBRUP recommended a clinical trial to demonstrate efficacy in the subgroup of women 35-42 years old. In addition, CMC deficiencies were issued in the Complete Response. The current NDA was resubmitted by Ferring on 02/25/2016 following the Complete Response issued on 2/28/2011 to address the CMC issues only. To address the CMC deficiencies, the (b) (4) method used in the drug product manufacturing process was replaced with a (b) (4) method. There were no changes to the formulation itself. The qualitative and quantitative composition of the 2 vaginal rings was identical. To demonstrate the BE between the “to-be-marketed” Progesterone Vaginal Rings manufactured using the new and legacy processes, Ferring submitted a new clinical BE study (Study # DR201-BE-10021). Therefore, this review is focused on the new BE study.

1.1 Recommendation

The Office of Clinical Pharmacology (OCP)/Division of Clinical Pharmacology 3 (DCP-3) has reviewed NDA 201110 resubmitted on February 25, 2016. The DR-201 progesterone vaginal ring produced from the new process was bioequivalent to the DR-201 progesterone vaginal ring produced from the legacy process. The overall Clinical Pharmacology information submitted to support this NDA is acceptable.

1.2 Post-marketing Requirements or Commitments

None

1.3 Summary of Important Clinical Pharmacology Findings

The subject of NDA 201110 is a progesterone-releasing, silicone vaginal ring formulation, intended for once-weekly administration over 10 weeks as part of luteal phase supplementation in infertile women undergoing Assisted Reproductive Technology (ART) treatment for achieving pregnancy. Each weekly vaginal ring consists of ~1.8 g of progesterone in a silicone matrix at a concentration of (b) (4).

In the re-submission dated 02/25/2016, the sponsor addressed a CMC deficiency by changing product manufacturing process and submitted the results of a comparative PK study to demonstrate the bioequivalence between the “to-be-marketed” progesterone vaginal rings manufactured using the new and legacy processes. The BE study (Study # DR-BE-10021) is an open-label, single-dose, randomized, 2-treatment, 2-period crossover study with a 7 days washout period between two treatments conducted in 56 healthy postmenopausal women. A summary of the PK parameters of this study is given below (Table 1). The inter-subject variability was ~ 33 % and intra-subject variability was ~ 25% in this study.

Table 1. A summary of progesterone PK in the BE study DR-BE-10021

Parameter	Treatment A (N=56)	Treatment B (N=56)
AUC _{0-∞} (ng•h/mL)	1140 (289)	1174 (382)
AUC _{0-t} (ng•h/mL)	1136 (289)	1169 (381)
C _{max} (ng/mL)	8.441 (2.216)	8.778 (2.475)

BE analyses were conducted on the baseline-corrected data. The point estimates for the ratio between products from the new process vs. legacy process for all three PK parameters, C_{max}, AUC_{0-t}, and AUC_{0-∞}, were between 96 - 100%, while the 90 % confidence intervals (CIs) surrounding these ratios were contained entirely within the BE limits of 80 - 125%. Thus the two products met the BE criteria. Similar results were seen for baseline uncorrected data.

Quantitation of progesterone in human plasma and tissue samples was achieved using validated LC-MS/MS methods. Validation study outcomes including precision and accuracy data and reports of the sample analyses were found to be acceptable.

2 QUESTION BASED REVIEW

2.1 General Attributes

Reviewed by Dr. Sandhya Apparaju dated 02/02/2011 and no new data was submitted.

2.2 General Clinical Pharmacology

Reviewed by Dr. Sandhya Apparaju dated 02/02/2011 and no new data was submitted.

2.3 Intrinsic Factors

Reviewed by Dr. Sandhya Apparaju dated 02/02/2011 and no new data was submitted.

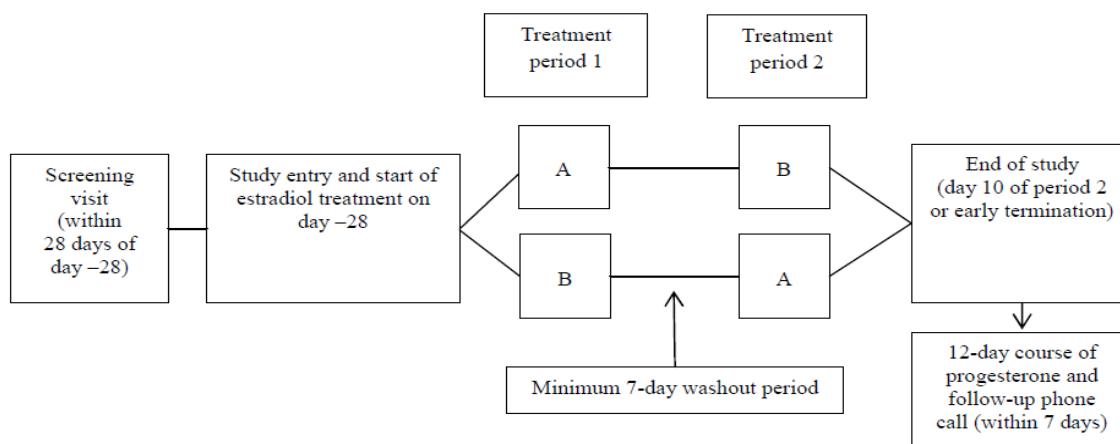
2.4 Extrinsic Factors

Reviewed by Dr. Sandhya Apparaju dated 02/02/2011 and no new data was submitted.

2.5 General Biopharmaceutics

2.5.1 Is the proposed to-be-marketed formulation produced from the new process bioequivalent to the to-be-marketed formulation of progesterone vaginal ring produced from the legacy process?

Yes, the proposed to-be-marketed formulation produced from the new process was demonstrated to be bioequivalent to the formulation produced from the legacy process as seen from the Study DR201-BE-10021 conducted in healthy post-menopausal females. The (b) (4) method used in the legacy manufacturing process was replaced with a (b) (4) method in the new process. There were no changes to the formulation itself.



A=Treatment A, test; DR-201 progesterone vaginal ring (b)(4)% (weight/weight [w/w]) from the new process, 1 ring inserted for 7 days.

B=Treatment B, reference; DR-201 progesterone vaginal ring (b)(4)% (w/w) from the legacy process, 1 ring inserted for 7 days.

Figure 1. The schema of the BE Study DR201-BE-10021 submitted on 02/25/2016

In the submissions dated 04/30/2010, Teva provided a comparative PK study report (Study # 10936035) to demonstrate BE between a clinical trial formulation and a to-be-market commercial formulation after a change in site of manufacturing only. On 02/25/2016, after a change in manufacturing process, the new sponsor of NDA 201110, Ferring submitted the second clinical BE study (Study # DR201-BE-10021) to demonstrate BE between the “to-be-marketed” Progesterone Vaginal Rings manufactured using a new process and a legacy process. The design of the Study # DR201-BE-10021 is shown in Figure 1. A comparison of study design between the two BE studies is summarized in Table 2. There are only two revisions in the Study DR201-BE-10021: (1) The subject number is increased from 32 to 56; (2) One addition blood sample collection time point (171 hour) is included in the PK study. Both revisions are acceptable.

Table 2. A comparison between prior BE Study 10936035 and current BE Study DR201-BE-10021

	Study 10936035	Study DR201-BE-10021
Study design	open-Label, single-dose, randomized, 2-treatment, 2-period crossover, 7 days washout	open-Label, single-dose, randomized, 2-treatment, 2-period crossover, 7 days washout
Subjects	32 healthy post-menopausal females	56 healthy post-menopausal females
Estradiol treatment	1 mg estradiol for at least 28 days prior to first dose and continued throughout the entire study, until the final PK blood draw on day 10 in treatment period 2	1 mg estradiol for at least 28 days prior to first dose and continued throughout the entire study, until the final PK blood draw on day 10 in treatment period 2
Blood sample collection	-1, -0.5, and 0 hours (within 10 minutes before ring insertion) and after vaginal ring insertion at 1, 2, 3, 4, 6, 8, 12, 24, 48, 72, 96, 120, 144, 168 (before vaginal ring removal), 169, 170, 172, 174, 176, 180, 192, and 216 hours	-1, -0.5, and 0 hours (within 10 minutes before ring insertion) and after vaginal ring insertion at 1, 2, 3, 4, 6, 8, 12, 24, 48, 72, 96, 120, 144, 168 (before vaginal ring removal), 169, 170, 171, 172, 174, 176, 180, 192, and 216 hours

The BE study design was an open-label, single dose (7-day vaginal ring), multiple site, two-treatment, two-period, two-sequence, randomized crossover study in N= 56 post-menopausal women. Subjects received a single dose of once-weekly vaginal ring formulation - either Reference (the formulation produced from the legacy process) or Test (the formulation produced from the new process) in period I or period II of the study separated by 7-day washout duration. All subjects received pre-treatment with 1 mg estradiol oral tablets for at least 28 days prior to first dose and also continued with this treatment throughout the entire study including the washout period, until after the final blood draw in period II.

PK sampling and analysis: Blood samples for plasma progesterone analysis were collected at pre-dose (for baseline; -1.0, -0.5 and 0 h), extensively during the two treatment periods and after removal of the vaginal ring in the second period. Samples were analyzed by (b) (4) using a validated LCMS/MS methodology. All post-dose samples were corrected for baseline by deducting the mean of three pre-dose values (-1, -0.5 and 0 hours). PK analyses were conducted on both baseline-corrected data non-baseline corrected data. Non-baseline corrected data analysis is presented as supportive data. BE was based on the 90 % confidence intervals surrounding treatment mean ratios (T: R) for AUC0-t, AUC0-inf and Cmax.

The baseline corrected progesterone plasma concentration - time profiles and non-baseline corrected progesterone plasma concentration - time profiles determined in Study DR201-BE-10021 are shown in Figure 2 and Figure 3, respectively. The progesterone plasma concentration - time profiles of test (Treatment A) and reference (Treatment B) formulations are comparable with baseline correction or without baseline correction. Visual inspection cannot detect significant difference between Treatment A and Treatment B. The baseline corrected progesterone plasma concentration - time profiles of test (commercial batch) and reference (clinical trial batch) determined in the Study 10936035 (Figure 4) are similar to that observed in the Study DR201-BE-10021 (Figure 2).

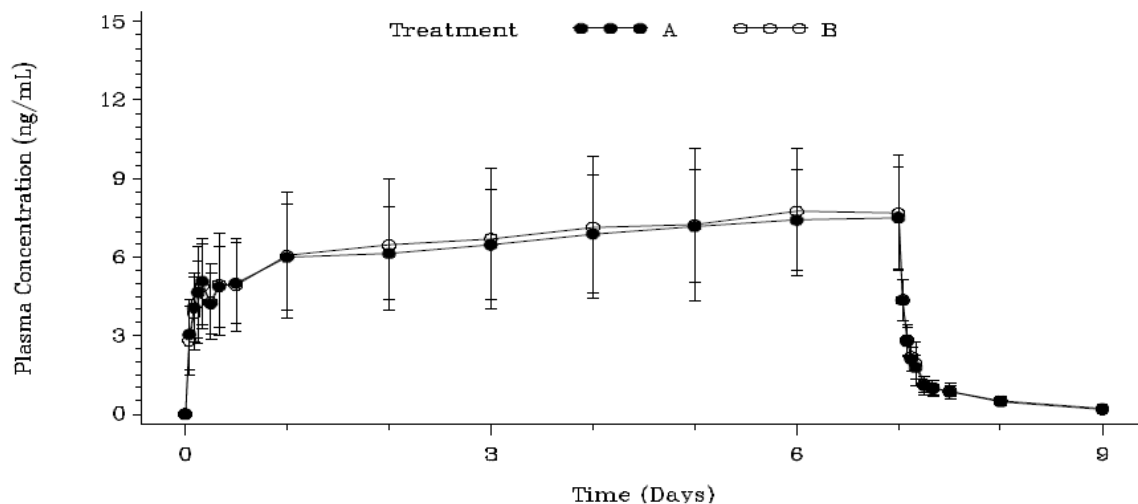


Figure 2. Baseline corrected progesterone plasma concentration - time profile in postmenopausal female subjects (BE Study DR201-BE-10021)
Treatment A: Formulation produced from the new process
Treatment B: Formulation produced from the legacy process

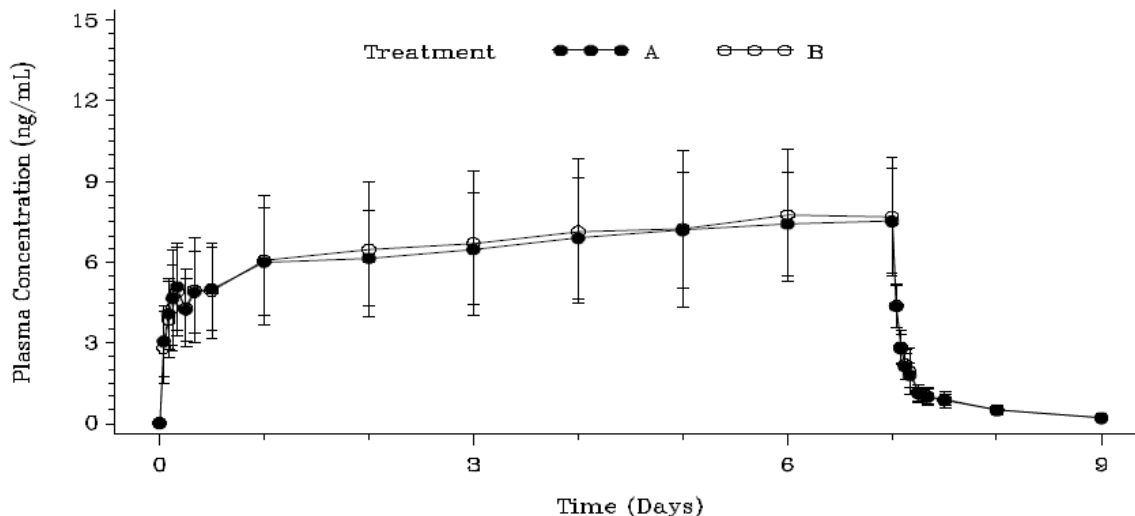


Figure 3. Non-baseline corrected progesterone plasma concentration - time profile in postmenopausal female subjects (BE Study DR201-BE-10021)

Treatment A: Formulation produced from the new process

Treatment B: Formulation produced from the legacy process

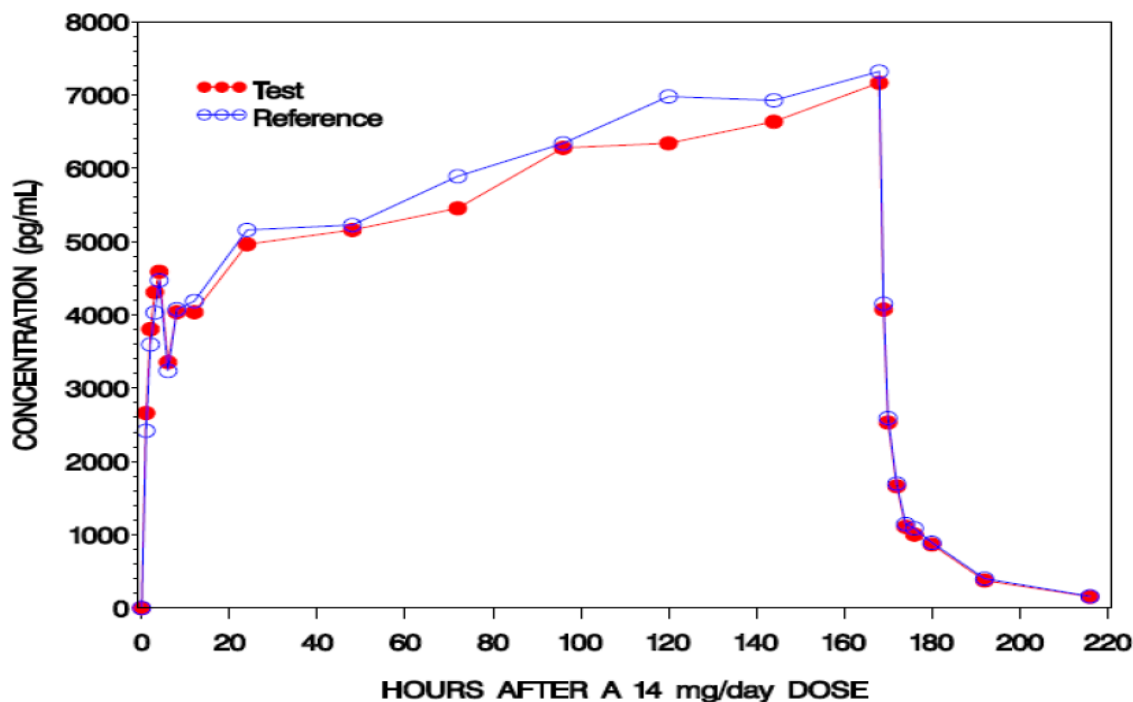


Figure 4. Baseline corrected progesterone plasma concentration - time profile in postmenopausal female subjects (from prior BE Study 10936035)

Test: Commercial batch; Reference: Clinical trial batch

Mean pharmacokinetic parameters of baseline corrected progesterone after administration of test and reference formulations in Study DR201-BE-10021 are summarized in Table 3. After administration of progesterone via Treatment A and Treatment B, the mean $AUC_{0-\infty}$ values for baseline corrected progesterone were 1140 and 1174 ng•h/mL, respectively. Comparable values were observed for AUC_{0-t} . Mean peak plasma concentrations (C_{max}) were 8.441 and 8.778 ng/mL,

respectively. Mean $t_{1/2}$ of progesterone was 16.41 and 16.75 hours, respectively. The median t_{max} of baseline corrected progesterone was achieved at 120 and 144 hours, respectively. All calculated pharmacokinetic parameters appeared to be comparable between Treatment A and Treatment B. Pharmacokinetic parameters for non-baseline corrected progesterone were similar to those observed for baseline-adjusted progesterone (Table 4).

Table 3. Baseline corrected progesterone PK parameters in the BE Study DR201-BE-10021
Treatment A: Formulation produced from the new process
Treatment B: Formulation produced from the legacy process

Parameter	Treatment A (N=56)	Treatment B (N=56)
AUC _{0-∞} (ng•h/mL)	1140 (289)	1174 (382)
AUC _{0-t} (ng•h/mL)	1136 (289)	1169 (381)
C _{max} (ng/mL)	8.441 (2.216)	8.778 (2.475)
$t_{1/2}$ (h)	16.41 (2.93)	16.75 (3.49)
t_{max} (h) ^a	120.00 (3.00, 167.87)	144.00 (6.00, 167.75)
λ_z (1/h)	0.0435 (0.0072)	0.0433 (0.0103)

Table 4. Non-baseline corrected progesterone PK parameters in the BE Study DR201-BE-10021

Treatment A: Formulation produced from the new process
Treatment B: Formulation produced from the legacy process

Parameter	Treatment A (N=56)	Treatment B (N=56)
AUC _{0-∞} (ng•h/mL)	1143 (290)	1177 (384)
AUC _{0-t} (ng•h/mL)	1138 (289)	1172 (383)
C _{max} (ng/mL)	8.452 (2.218)	8.791 (2.486)
$t_{1/2}$ (h)	16.92 (2.99)	17.36 (3.15)
t_{max} (h) ^a	120.00 (3.00, 167.87)	144.00 (6.00, 167.75)
λ_z (1/h)	0.0421 (0.0068)	0.0411 (0.0070)

As a comparison, the mean pharmacokinetic parameters of baseline-corrected progesterone determined in the Study 10936035 are summarized in Table 5. The mean values of C_{max}, AUC_{0-t}, and AUC_{0-∞} determined in the BE Study 10936035 are approximately 10% lower than the corresponding values determined in the BE Study DR201-BE-10021. Considering the 28-33% inter-subject variability in PK parameters and inter-laboratory variability in sample preparation and plasma progesterone measurement, the 10% difference in the mean values of C_{max}, AUC_{0-t}, and AUC_{0-∞} between the two BE studies is acceptable.

Table 5. Baseline corrected progesterone PK parameters in prior BE Study 10936035 (N = 32)

PK Parameter	Test A (Commercial batch)	Reference B (Clinical trial batch)
AUC _{0-t} (ng.h/mL)	1005 ± 315 (31 %)	1045 ± 283 (27 %)
AUC _{0-inf} (ng.h/mL)	1009 ± 316 (31 %)	1049 ± 283 (27 %)
C _{max} (ng/mL)	7.5 ± 2.3 (30 %)	7.8 ± 2.0 (26 %)
T _{max} (h)	146 ± 33 (12 %)	143 ± 37 (23 %)
T _{1/2} (h)	14.5 ± 1.7 (12 %)	14.8 ± 2.4 (15.7 %)

As shown in Table 6, the upper and lower bounds of the 90% CIs of the ratio of geometric least square (LS) means (Treatment A/Treatment B) for C_{max}, AUC_{0-t}, and AUC_{0-∞} for baseline corrected progesterone were fully contained within the BE boundaries of 0.80-1.25. Results from the non-baseline corrected statistical analysis were consistent with results of the baseline corrected analysis (Table 7). The inter-subject variability was ~ 33 % and intra-subject variability was ~ 25 % in the Study DR201-BE-10021.

Table 6. BE analysis of baseline corrected progesterone C_{max}, AUC_{0-t}, and AUC_{0-∞} in postmenopausal female subjects

Treatment A: Formulation produced from the new process

Treatment B: Formulation produced from the legacy process

Parameter	Treatment	N	Geometric LS means	Ratio of geometric LS means (A/B)	90% CI of ratio (A/B)	Root MSE	Intra-subject CV
C _{max} (ng/mL)	A	56	8.166	0.967	0.9263, 1.0102	0.137	13.765
	B	56	8.442				
AUC _{0-t} (ng.h/mL)	A	56	1101.701	0.999	0.9430, 1.0575	0.181	18.267
	B	56	1103.257				
AUC _{0-∞} (ng.h/mL)	A	56	1106.188	0.998	0.9427, 1.0571	0.181	18.244
	B	56	1108.106				

Table 7. BE analysis of non-baseline corrected progesterone C_{max}, AUC_{0-t}, and AUC_{0-∞} in postmenopausal female subjects.

Treatment A: Formulation produced from the new process

Treatment B: Formulation produced from the legacy process

Parameter	Treatment	N	Geometric LS means	Ratio of geometric LS means (A/B)	90% CI of ratio (A/B)	Root MSE	Intra-subject CV
C _{max} (ng/mL)	A	56	8.176	0.967	0.9263, 1.0103	0.137	13.786
	B	56	8.452				
AUC _{0-t} (ng.h/mL)	A	56	1104.056	0.999	0.9431, 1.0574	0.181	18.236
	B	56	1105.583				
AUC _{0-∞} (ng.h/mL)	A	56	1108.935	0.998	0.9429, 1.0570	0.181	18.206
	B	56	1110.805				

Certain aspects of the BE study design were adequately planned to account for potential confounding issues such as the following. Firstly, all study subjects (post-menopausal women) were dosed with oral estrogen for 28 days prior to period 1 (and continued throughout the study) to normalize any differences in vaginal atrophy which is more prevalent in this population. In addition, the sponsor also allowed 7-day wash-out duration between the two study periods, thereby avoiding concerns related to drug-carryover and potentially pharmacological effect carryover into the second period. In addition, since the current study was a 2-sequence crossover study, half of the subjects received treatment A (test) followed by treatment B (reference) while the other half received B followed by A. In essence any change that might favor B over A in one sequence will favor A over B in the other sequence and cancel out of the treatment comparison. Desirable features of the study design such as 2-way cross-over of treatments, use of an adequate wash-out duration between test and reference, and use of pre-treatment with estradiol suggest acceptability of the overall information presented. Hence the study is acceptable in establishing BE of two to-be-marketed formulations of the DR-201 vaginal ring produced from the new process and the legacy process.

2.5.2 What are the safety or efficacy issues, if any, for BE studies?

The safety data indicate that administration of progesterone (with a release rate of approximately 11 mg/day for up to 7 days) via a 7-day application cycle of the to-be-marketed vaginal ring produced from Treatment A and Treatment B were similarly safe and well tolerated in healthy postmenopausal female subjects who were pretreated with estrogen. No deaths, other serious adverse events, withdrawals due to adverse events, or severe adverse events occurred in this study. The incidence of mild and moderate adverse events after subjects received either Treatment A or Treatment B was similar. The majority of subjects after receiving either Treatment A or Treatment B (89% and 86%, respectively) had at least 1 adverse event. The most frequent adverse event for each treatment was withdrawal bleed (70% for Treatment A and 73% for Treatment B). Other frequent adverse events included pelvic pain (14% for Treatment A and 16% for Treatment B), vaginal discharge (20% for Treatment A and 11% for Treatment B), and breast tenderness (13% for Treatment A and 7% for Treatment B). Given the relatively small sample size, the adverse event profiles were not notably different between Treatment A and Treatment B.

Although shifts in several serum chemistry and hematology parameters occurred from the reference range at baseline to outside the reference range at endpoint, none were clinically meaningful, and there were no trends in any clinical laboratory variable. A total of 19 (34%) subjects had potentially clinically significant urinalysis abnormalities (primarily a ≥ 2 -unit increase from baseline in blood in the urine).

There were no notable changes or adverse events regarding vital signs, ECG assessments, or physical examination findings for subjects administered either Treatment A or Treatment B, except for 1 subject who had a transient adverse event of dermatitis contact as a physical examination finding.

None of the concomitant medications had a negative impact on safety.

The safety data indicate that Treatment A and Treatment B were similarly safe and well tolerated by healthy postmenopausal female subjects who were pretreated with estrogen.

2.5.3 What were the site inspection findings for BE study DR201-BE-10021?

On 03/30/2016, the Office of Clinical Pharmacology (OCP)/Division of Clinical Pharmacology 3 (DCP-3) sent a request to the Office of Study Integrity and Surveillance (OSIS) to inspect the clinical site (SeaView Research Inc., Miami, FL) and analytical site (b) (4) of the new BE study # DR201-BE-10021¹. The inspection reports for the clinical site² and analytical site³ were uploaded to DARRTS on 06/30/2016 and 07/01/2016, respectively. The inspection of the clinical site of the study was conducted by ORA (FLA-DO) investigator Angelica M. Chica at SeaView Research, Inc., Miami, FL from May 9–24, 2016. No significant issues were observed and no Form FDA 483 was issued. The OSIS reviewer concluded that the clinical data from the audited study at SeaView Research, Inc. were reliable. For the analytical site at (b) (4) OSIS recommended accepting data without an on-site inspection because the site was recently inspected by OSIS and the inspectional outcome from the inspection was classified as No Action Indicated (NAI). According to the OSIS inspection reports, the data from Study DR201-BE-10021 submitted to NDA 201110 are acceptable for further Agency review.

2.6 Bioanalytical Methods

Quantitation of progesterone in human plasma from this study was conducted using a validated HPLC-MS/MS method. The method validation and sample analyses were conducted by (b) (4). The method validation data was reviewed by Dr. Sandhya Apparaju on 02/02/2011 and found acceptable. There was no change in the method used for the BE study DR-BE-10021.

For the BE study DR-BE-10021, analysis of human plasma samples began on 25 April 2014 and was completed on 19 June 2014. Totally, 2909 human plasma samples were analyzed using the validated method. Precision and accuracy were evaluated by replicate analyses of surrogate matrix quality control pools (QCs 1, 3, and 5) prepared at three concentrations spanning the calibration range. Additional mid-low and mid-high level quality controls (QCs 2 and 4) were also prepared in human plasma containing dipotassium EDTA. Precision was measured as the percent coefficient of variation (%C.V.) of the set of values for each pool. Accuracy was expressed as the percent difference of the mean value for each pool from the theoretical concentration. Inter-assay data are presented below.

¹ OSIS Consult Request for Biopharmaceutical Inspections by (b) (4)

² Bioequivalence Establishment Inspection Report Review by Yiyue Zhang dated 06/30/2016

³ Bioequivalence Establishment Inspection Report Review by Shila S. Nkah dated 07/01/2016

Inter-assay Precision and Accuracy

Run ID	QC 1 (ng/mL)	QC 2 (ng/mL)	QC 3 (ng/mL)	QC 4 (ng/mL)	QC 5 (ng/mL)
25RCAQ-A	0.162	0.473	1.488	6.008	35.564
	0.152	0.481	1.602	6.407	39.446
26RCAQ-A	0.165	0.446	1.434	6.174	38.245
	0.146	0.510	1.469	6.372	36.940
N	46	46	46	46	46
Theoretical Concentration	0.150	0.474	1.500	6.390	37.500
Mean	0.150	0.467	1.501	6.268	37.722
S.D.	0.012	0.027	0.096	0.366	2.761
%C.V.	7.8	5.9	6.4	5.8	7.3
% Difference from Theoretical	0.0	-1.5	0.1	-1.9	0.6
Low Limit	0.128	0.403	1.275	5.432	31.875
High Limit	0.173	0.545	1.725	7.349	43.125

QCs 1, 3, and 5 were prepared in surrogate matrix.

QCs 2 and 4 were prepared in human plasma.

To demonstrate reproducible quantitation of incurred subject samples, approximately 10% of the study samples were reassayed. The incurred sample reproducibility (ISR) values were used for comparison purposes and are included in the analytical report but not used in determining the final reported value. Incurred sample repeats were considered acceptable if the original and reassay values from two-thirds of the repeated samples had a relative percent difference of $\leq$ (b) (4)%. The results of the incurred sample repeats met the acceptance criteria.

Overall, plasma sample analysis was acceptable.

3 Detailed Labeling Recommendations

The following changes are recommended in the Sponsor's proposed labeling. The double-underlined text indicates insertion recommended by the reviewer and the ~~strikethrough~~ text indicates recommended deletion.

Clinical Pharmacology's recommendation in prior review dated 02/02/2011

DOSAGE FORMS AND STRENGTHS

(b) (4) 1.78 g progesterone, (b) (4) ~~an average of 11~~
~~mg/day progesterone over a 7-day period~~ (b) (4) [3]

Sponsor's draft labeling dated 02/25/2016:

Vaginal (b) (4) 1.78 g progesterone, (b) (4) an average of 11 mg/day
progesterone over a 7-day period (b) (4) [3]

Sponsor's justification: Ferring prefers not to add (b) (4) as per the General Advice Letter dated March 4, 2011 (Reference ID 2913939). Ferring's position is that this

confidential proprietary formulation info would be confusing to prescribers, and is not the convention for other vaginal rings on the market.

Reviewer's comment: The sponsor's justification is acceptable.

Clinical Pharmacology's recommendation in prior review dated 02/02/2011

3. DOSAGE FORMS AND STRENGTHS

(b) (4) a white to off-white, flexible, non-biodegradable, (b) (4) silicone vaginal ring (b) (4) a cross-sectional diameter of (b) (4) mm and an outer diameter of 55 mm.

Sponsor's draft labeling dated 02/25/2016:

(b) (4) a white to off-white, flexible, non-biodegradable, (b) (4) silicone (b) (4) ring (b) (4) a cross-sectional diameter of (b) (4) mm and an outer diameter of 55 mm.

Sponsor's justification: Ferring prefers not to add (b) (4) as per the General Advice Letter dated March 4, 2011 (Reference ID 2913939). Ferring's position is that this confidential proprietary formulation info would be confusing to prescribers, and is not the convention for other vaginal rings on the market.

Reviewer's comment: The sponsor's justification is acceptable.

Clinical Pharmacology's recommendation in prior review dated 02/02/2011

8. USE IN SPECIFIC POPULATIONS

8.6. Body mass index (BMI)

Efficacy of [TRADENAME] in women with a BMI > 38 kg/m² has not been studied.

Reviewer's comment: The sponsor accepted clinical pharmacology's edits.

Clinical Pharmacology's recommendation in prior review dated 02/02/2011

11. DESCRIPTION

When placed in the vagina, each (b) (4) is estimated to provide an average release rate of 11 mg/day of progesterone (b) (4) over 7 days (b) (4).

Reviewer's comment: The sponsor accepted clinical pharmacology's edits.

Clinical Pharmacology's recommendation in prior review dated 02/02/2011

12.3. Pharmacokinetics

Reviewer's comment: The sponsor accepted clinical pharmacology's edits.

Overall, the sponsor has accepted most labeling revisions recommended by the clinical pharmacology reviewer in the prior review dated 02/02/2011. The sponsor declined to add (b) (4) per clinical pharmacology reviewer's recommendation because the sponsor considered it as confidential proprietary formulation information. The sponsor's justification is acceptable.

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/s/

PENG ZOU
09/07/2016

DOANH C TRAN
09/07/2016

OFFICE OF CLINICAL PHARMACOLOGY REVIEW

NDA: 201110	Submission Dates: 04/30/2010, 08/13/2010
Brand Name	[Pending]
Generic Name	Progesterone
Reviewer	Sandhya Apparaju, Ph.D.
Team Leader	Myong Jin Kim, Pharm.D.
OCP Division	Division of Clinical Pharmacology 3
OND Division	Division of Reproductive and Urologic Products
Sponsor	Teva Women's Health R & D
Relevant IND(s)	70,875
Submission Type	Original NDA; Standard
Formulation; Strength(s)	Vaginal ring for once-weekly use; (b) (4)
Indication	Progesterone supplementation for up to 10 weeks post-embryo transfer as part of Assisted Reproductive Technology (ART) treatment in infertile women

An optional intra-divisional level briefing was held for this NDA on Dec 15, 2010 in Room 3300 of WO BLDG 51. Attendees included Dr's. Hae Young Ahn, Myong Jin Kim, Christos Mastroyannis, Hyunjin Kim, Chongwoo Yu, Sayed Al Habet, Julia Cho, Li Li and Sandhya Apparaju.

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1 Executive Summary

1.1 Recommendation

NDA 201110 is acceptable from a Clinical Pharmacology perspective provided that a satisfactory agreement is reached between the sponsor and the Agency regarding language in the package insert.

1.2 Phase IV Commitments

None

1.3 Summary of Important Clinical Pharmacology and Biopharmaceutics Findings

The subject of NDA 201110 is a progesterone-releasing, silicone vaginal ring formulation, intended for once-weekly administration over 10 weeks as part of luteal phase supplementation in infertile women undergoing Assisted Reproductive Technology (ART) treatment for achieving pregnancy. The drug product has been developed by the Teva Women's Health Division (formerly Duramed Research, Inc.) of Teva Global Branded Pharmaceuticals R & D, Inc. Each weekly vaginal ring consists of ~1.8 g of progesterone in a silicone matrix at a concentration of (b) (4).

Sponsor has conducted 7 phase 1, 2 & 3 clinical trials in order to assess progesterone pharmacokinetics (PK; systemic and tissue progesterone), pharmacodynamics (PD; successful secretory transformation of the endometrium) and efficacy (clinical pregnancy rates) with the progesterone vaginal ring formulation. In addition, bioequivalence of the clinical vs. commercial formulations (change in site of manufacturing only) and estimation of in vivo drug release have been addressed.

The PK of progesterone from the proposed vaginal ring formulation were addressed in post-menopausal healthy volunteers during the definitive PK study 10716222 following single dose and steady state dosing. Mean progesterone concentration-time profiles are shown following the two doses:

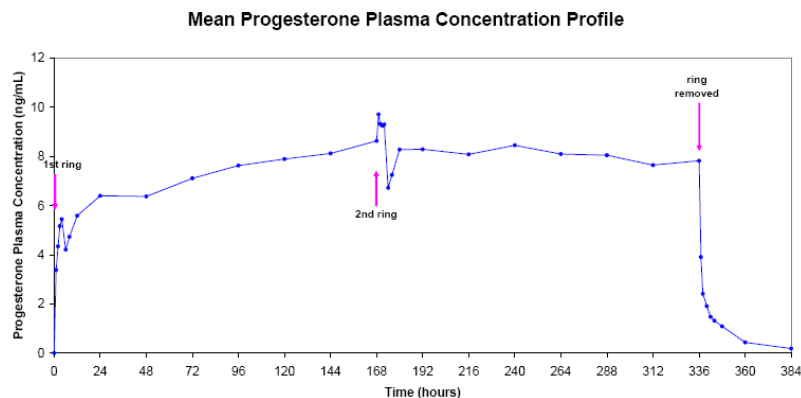


Figure 1: Average plasma progesterone concentration vs. time profiles following two consecutive weekly doses of progesterone vaginal ring formulation in post-menopausal volunteers of study 10716222.

Cmax values after the first dose and second doses of the vaginal ring averaged at 9.3 ng/mL and 10.7 ng/mL, respectively. The average value of steady-state plasma progesterone (Cavg) observed during the second dose of the ring was 8.2 ng/mL. The inter-individual variability (% CV) in plasma progesterone Cmax and AUC following the vaginal ring product was ~25- 30 %.

Serum progesterone concentrations were measured using standard clinical chemistry methods in the phase 2 and phase 3 clinical trials in the target population (infertile premenopausal women undergoing ART). In women who became pregnant during the ART procedures, the endogenous progesterone concentrations continued to increase markedly thus confounding the contribution from exogenous progesterone.

Only a single dose strength of the vaginal ring (weekly) formulation was evaluated throughout this NDA [(b) (4) containing ~1.8 g of progesterone per ring]. Therefore, dose-response information for safety and efficacy is not available. Data from phase 2 studies is suggestive of a trend for increasing endometrial transformation rates (PD) with increasing serum progesterone concentrations. However, at this time there is no established correlation between this PD measure and the primary efficacy endpoint (i.e. clinical pregnancy rates). There is also no established correlation between serum progesterone concentrations and efficacy.

Study 10936035 evaluated the bioequivalence of the clinical trial formulation of the progesterone vaginal ring ((b) (4)) to that of the proposed commercial formulation ((b) (4)) in post menopausal female volunteers. The two formulations differed only in the site of manufacturing. Baseline correction of progesterone concentrations was done by subtracting the mean of three pre-dose values (-1, -0.5 and 0 hours) from all post-dose samples.

PK analyses were conducted on the baseline-corrected data. The results of the BE analyses are shown below. The point estimates (expressed as % reference B in the table) for all three PK parameters, Cmax, AUCt, and AUCinf, were ~ 95 %, while the 90 % confidence intervals (CIs) surrounding these ratios were contained entirely within the bioequivalence limits of 80-125 %. Thus the two treatments met the bioequivalence criteria. The inter-subject variability was ~ 25 % and intra-subject variability was ~ 13 % in this study. Similar data was seen for both baseline corrected and uncorrected data.

Table 1: Results of the bioequivalence analysis [Study 10936035]

	Test	Reference	Ratio [% Ref]	90 % CI range
Cmax (ng/mL)	7.14	7.54	94.62	89.37 - 100.18
AUC (ng h/mL)	949.35	1002.21	94.73	89.51 - 100.25
AUCinf (ng h/mL)	952.53	1005.53	94.73	89.52 - 100.24

Residual drug content was determined and apparent in vivo release rates were estimated using used vaginal rings and a similar number of unused control rings from the same batch following the BE study. Data suggests an average in vivo release rate of (b) (4) mg per day from the vaginal ring (obtained as mg of progesterone released over one week, divided by 7).

Quantitation of progesterone in human plasma and tissue samples was achieved using validated LC-MS/MS methods. Validation study outcomes including precision and accuracy data and reports of the sample analyses were found to be acceptable.

No specific population studies were conducted in support of this NDA. Phase 3 exclusions included patients with significant disease including renal or hepatic impairment. A trend towards decreasing drug concentrations with increasing body mass index (BMI) was noted in the BE study but not in the definitive PK study (upper limit of BMI for enrollment was 32 kg/m² in both studies). A trend for lower efficacy (clinical pregnancy rates) with increasing BMI and with increasing age was also noted in the phase 3 clinical trial (upper limits for enrollment in phase 3 were 38 kg/m² for BMI and 42 years for age). However, the trial was not powered to evaluate differences in subgroups. The labeling for this product should indicate that the phase 3 trial did not include patients with BMI > 38 kg/m² and that the PK in presence of renal or hepatic impairment have not been assessed.

No specific drug-drug interaction studies were conducted for the progesterone vaginal ring formulation. The phase 3 clinical trial did not allow concomitant use of other vaginal products such as anti-fungals while using the study drug. Since the possibility of a potential effect of other vaginal formulations on the release and/or absorption from the vaginal ring product is not known, the labeling should indicate that concomitant vaginal formulations should be avoided while on this drug (for up to 10 weeks).

2 Question Based Review

2.1 General Attributes of the Drug

- 2.1.1 What are the highlights of the chemistry and physical-chemical properties of the drug substance and the formulation of the drug product as they relate to clinical pharmacology and biopharmaceutics review?

Drug substance: Progesterone (pregn-4-ene-3,20-dione) is a steroid hormone with the following structure:

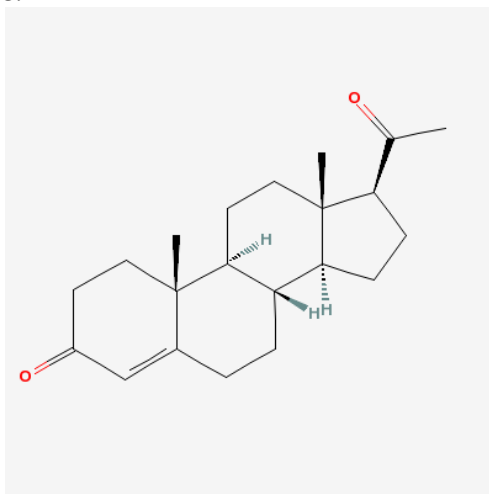


Figure 2: Chemical structure of progesterone

The molecular formula of Progesterone is $C_{21}H_{30}O_2$. The molecular mass of Progesterone is 314.47. General physicochemical properties are given below:

Physical Description: White to practically white powder

Solubility in ethanol (at room temperature, RT): freely soluble, 1 in 8 of ethanol

Solubility in acetone (at RT): sparingly soluble

Solubility in water (at RT): practically insoluble

Melting Range: 126°C to 131°C

Drug Product: The proposed progesterone vaginal ring is a non-biodegradable, white to off-white, flexible silicone ring containing Progesterone as the active ingredient. The vaginal ring consists of (b) (4) light mineral oil and Progesterone, USP (b) (4). The active ingredient is present in (b) (4) concentration dispersed evenly within the ring. Each Progesterone Vaginal ring contains approximately 1.8 grams of USP grade progesterone.

Table 2: The unit formula for proposed progesterone vaginal ring formulation

Ingredient	Composition Per Ring	
	Weight, g	Percentage, w/w
Progesterone, USP (b) (4)	1.7800	(b) (4)
Light Mineral Oil (b) (4)	(b) (4)	(b) (4)
Total per Ring		

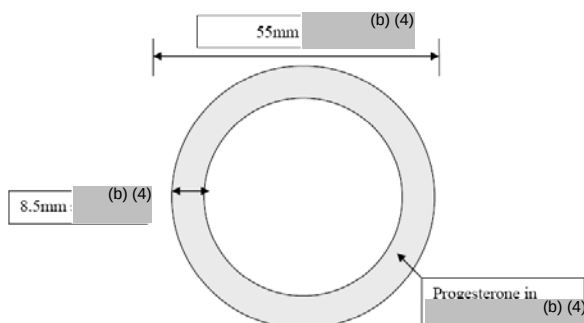


Figure 3: Progesterone vaginal ring dimensions

Data obtained via residual drug content analysis in used rings suggests that when placed in the vagina, each ring has an average release rate of 11 mg/day of progesterone over the dosing interval.

2.1.2 What is the proposed mechanism of action and therapeutic indication?

Indication: The proposed progesterone vaginal ring formulation is indicated to support embryo implantation and early pregnancy (up to 10 weeks post-embryo transfer) by supplementation of corpus luteal function as part of an ART treatment program for infertile women.

[ART involves the surgical removal of eggs from a woman's ovaries, fertilizing them with sperm in the laboratory, and then returning them to either the donor woman's or another woman's uterus].

Mechanism of action: Progesterone is a naturally occurring steroid that is secreted by the ovary, placenta, and adrenal gland. (b) (4)

(b) (4) In the presence of adequate estrogen, progesterone transforms a proliferative endometrium into a secretory endometrium. Progesterone is necessary to increase endometrial receptivity for implantation of an embryo. Once an embryo is implanted, progesterone acts to maintain a pregnancy.

Rationale for treatment: In women undergoing ART, use of a gonadotropin releasing hormone (GnRH) agonist as part of the regimen causes a temporary suppression of pituitary luteinizing hormone (LH) secretion. Without the LH signal, the corpus luteum may be dysfunctional, and subsequent progesterone and estrogen secretion may be abnormal, compromising endometrial receptivity, and potentially leading to decreased implantation and pregnancy rates. The goal of progesterone supplementation in ART is therefore to assist a corpus luteum that may have become compromised during ovulation induction or oocyte retrieval. Most treatment protocols advocate the use of progesterone throughout the first trimester of pregnancy, since corpus luteum activity has been demonstrated up to week 10 of pregnancy.

2.1.3 What is the proposed dosage and route of administration?

The proposed dose is one vaginal ring inserted vaginally starting the day after oocyte retrieval. The vaginal ring is replaced weekly, continuing for up to 10 weeks total duration if pregnancy occurs.

2.2 General Clinical Pharmacology

2.2.1 What are the design features of the clinical pharmacology and clinical studies used to support dosing or claims?

Sponsor has conducted 7 clinical trials in order to assess progesterone PK (systemic and tissue progesterone), PD (successful secretory transformation of the endometrium) and efficacy (clinical pregnancy rates) with the progesterone vaginal ring formulation. A separate report has been included to address the issue of residual drug content and estimated in vivo drug release.

Study populations evaluated: Phase 1 studies with the vaginal ring [definitive PK and BE studies] were generally conducted in post-menopausal women pre-treated with estrogen, in order to avoid the confounding influences of higher endogenous progesterone concentrations in pre-menopausal women. In addition, treating these postmenopausal women with estradiol for several days prior to insertion of the vaginal ring is expected to restore their vaginal epithelia to a substantially premenopausal state, which ensured that drug absorption from the rings would be comparable to that in the target ART population.

Previous discussions with the sponsor in this regard are captured under IND 70,875 in DARRTS, where the population proposed for these phase 1 studies (i.e. post-menopausal) though not similar to the target population (pre-menopausal women undergoing ART) was deemed acceptable for the purpose of characterizing PK and comparing formulations. PD endpoint (i.e. endometrial transformation) was assessed in the target pre-menopausal population.

The study design features for each of these clinical trials are briefly summarized:

Study 10716222 was a phase 1, open-label, single arm, single dose and multiple PK study of the proposed progesterone vaginal ring formulation in healthy post-menopausal volunteers (n =30).

Study 10936035 was a phase 1, randomized, blinded, crossover trial in 31 healthy post-menopausal volunteers evaluating the BE of the proposed commercial formulation (manufactured at the sponsor's OH commercial facility) to that of the phase 3 clinical trial formulation (manufactured at sponsor's NJ R & D facility). There were no differences in the actual formulation or the equipment used across these two sites.

Study 10936057 was used to obtain used progesterone vaginal rings for assessment of residual drug content and calculation of in vivo drug release.

Study DR-201-102 was a phase 1, randomized, open-label, active control, vs. no-treatment control trial to assess endometrial tissue progesterone, systemic progesterone, and endometrial PD outcomes (% patients achieving successful secretory transformation of the endometrium). Study was conducted in 63 healthy females 18-40 years with regular menstrual cycles.

Study DR-PGN-201 was a phase 2, randomized, open-label, active control study in 21 females aged 18-50 years who are candidates for donor oocyte program; the study was a 'mock cycle' preceding study DR-PGN-202 (below) and served to evaluate the proportion of subjects with adequate endometrial transformation.

Study DR-PGN-202 was a phase 2, randomized, open-label, active-control study in 9 participants who had demonstrated successful endometrial transformation in earlier study DR-PGN-201. Study assessed the clinical pregnancy rates (at 8 and 12 weeks of pregnancy);

Study DR-PGN-302 was a phase 3, randomized, investigator-blinded, active control study in 1399 (randomized) [594 completed] pre-menopausal women age 18-42 years with infertility, who are seeking ART. Study evaluated clinical pregnancy rates at 8 and 12 weeks of pregnancy as well as safety of the treatments.

2.2.2 What is the basis for selecting the response endpoints (i.e., clinical or surrogate endpoints) or biomarkers (collectively called PD) and how are they measured in clinical pharmacology and clinical studies?

Tissue progesterone concentrations in the endometrium were assessed as one of the PD outcomes in this NDA [Study DR-201-102]. This was intended to assess the 'local' progesterone exposure from the vaginal formulation. The PD outcome of most interest however, was the successful secretory transformation of the endometrium. Evidence of a secretory endometrium is accepted as an indicator of

adequate progesterone availability at the uterine site and has been used to support the development of other progesterone supplementation products. However, the correlation of either tissue progesterone or endometrial transformation with that of the primary efficacy outcome (i.e. clinical pregnancy) has not been established.

The accepted primary efficacy outcome for progesterone supplementation in ART is the 'rate of clinical pregnancy rates at 8 weeks (6 weeks after egg retrieval) or 12 weeks (10 weeks after egg retrieval)'. This was the primary outcome evaluated in this NDA as well. A two-side 95% CI was calculated around the difference in pregnancy rates between the new vaginal ring formulation and Crinone® (active comparator; approved vaginal gel). The non-inferiority of progesterone vaginal ring to Crinone® would be supported if the lower bound of that confidence interval exceeded -10%.

2.2.3 Are the active moieties in the plasma appropriately identified and measured to assess PK parameters and exposure response relationships?

Yes, the active moiety (i.e. progesterone) has been appropriately measured in biological matrices (plasma, serum and tissue) in this NDA. In the two phase 1 studies 10716222 (definitive PK) and 10936035 (BE study), plasma concentrations of progesterone were assessed using validated LC-MS/MS methods. These two studies provide the primary evidence of drug bioavailability, PK and formulation bioequivalence in this NDA. Quantitation of progesterone in human endometrium tissue was achieved in study DR-201-102 using HPLC method with MS/MS detection. Method validation and assay reports have been included in the NDA. Refer to the Bioanalytical section of this review for additional details. In the remaining phase 2 and phase 3 clinical trials, progesterone concentrations were periodically assessed in serum using chemiluminescent immunoassay methods. Plasma and serum matrices differ in that plasma contains fibrinogen and other clotting factors not present in serum. Progesterone binds principally to albumin (80%) and corticosteroid-binding globulin (17%). Since progesterone does not bind to fibrinogens, progesterone concentrations measured in plasma or serum using validated analytical methods should be comparable.

2.2.4 Exposure-response

2.2.4.1 What are the characteristics of the exposure-response relationships (dose-response, concentration-response) for efficacy and safety?

Only a single dose strength of the vaginal ring (weekly) formulation was evaluated throughout this NDA [(b) (4) containing ~1.8 g of progesterone per ring]. Therefore, dose-response information for safety and efficacy is not available.

Phase 2 information contained in the NDA is suggestive of a trend for increasing endometrial transformation rates (PD) with increasing serum progesterone concentrations. However, at this time there is no established correlation between endometrial transformation and clinical pregnancy rates. There is also no established correlation between serum progesterone concentrations and the primary efficacy outcomes i.e. clinical pregnancy.

Available serum progesterone and PD information from study DR-201-102 (employing the phase 3 clinical trial batch of the formulation) is summarized here:

- Study DR-201-102: In this study tissue and serum progesterone concentrations as well as endometrial transformation (PD) were assessed in 63 women who were either healthy pre-menopausal volunteers or those who were planning to undergo embryo transfer procedure. After screening, all women started transdermal estrogen therapy on day 1 of their menstrual cycle; sponsor notes that subjects underwent this ovarian suppression with pharmacologic doses of estrogen to minimize as much as possible the impact of endogenous ovarian hormonal production on serum and tissue progesterone measurements. After 14 days, subjects received one of three treatments below for 11-12 days; treatment with transdermal estradiol was continued in all subjects throughout the study period.
- Subjects were randomized to one of the three treatments: 1) transdermal estrogen only, 2) weekly progesterone vaginal ring (b) (4) and 3) Crinone vaginal gel 8 % 90 mg once daily. The treatment duration was 11-12 days.
- At the end of the treatment period, progesterone concentrations in the endometrial tissue were higher in the two active treatment groups (52.48 ng/g and 1437 ng/g in the vaginal ring and Crinone vaginal gel groups, respectively) compared to estrogen only group (2.82 ng/g).

Table 3: Tissue progesterone concentrations across treatments in study DR-201-102

Treatment	Tissue progesterone based on sample weight (ng/g)	Tissue progesterone – based on protein content (mg/g protein)
Estrogen only (n=21)	2.8 ± 8.2	0.11 ± 0.34
Vaginal ring (n=21)	52.5 ± 76.9	1.4 ± 2.12
Crinone vaginal gel (n= 21)	1437 ± 2370	34.7 ± 54.1

- Tissue concentrations in the Crinone vaginal gel group were markedly higher in several individuals of that group, although serum progesterone concentrations in this group were comparable to that of vaginal ring at the end of the treatment period (i.e. around the time of biopsy sampling; see table below). Sponsor attributes high tissue concentrations in the vaginal gel group to sampling error wherein the bioadhesive, viscous progesterone gel retained in the vagina and near the cervix could've been inadvertently

sampled while accessing the endometrium via the cervix for biopsy samples.

- The average serum progesterone concentrations at the end of the study were 7.9 and 9.3 ng/mL, for the vaginal ring and Crinone groups respectively.

Table 4: Serum progesterone concentrations across treatments in study DR-201-102

Serum progesterone (ng/mL)	Screening visit	Day 1 of treatment	Day 7 of treatment	End of treatment (11-12 days)
Estrogen only (n=21)	7.2 ± 7.1	1.47 ± 0.37	1.99 ± 2.8	1.4 ± 0.8
Vaginal ring (n=21)	5 ± 5.7	1.39 ± 0.27	7.06 ± 1.81	7.89 ± 2.36
Crinone gel (n=21)	4.1 ± 3.0	1.39 ± 0.29	10.1 ± 3.5	9.28 ± 5.15

[Note: Pre-treatment with transdermal estrogen patch reduces endogenous hormone levels; thus compared to the screening visit, serum progesterone levels at pre-dose on day 1 of treatment (14 days after initiation of pre-treatment with estrogen patch) are lower].

- Histology results indicate a clearly high incidence of endometrial transformation with the two progesterone groups at the end of treatment compared to estrogen alone group. The rate of transformation was somewhat higher with the vaginal ring group compared to the vaginal gel.

Table 5: Histology results (endometrial transformation status; PD) in study DR-201-102

Endometrial status classification	Estrogen only (n=19)	Vaginal ring (n=19)	Crinone vaginal gel 8 % (n=19)
Secretory	5 %	94.73 %	84.2 %
Other*	95 %	5.27 %	15.78 %

* proliferative, inactive, simple hyperplasia, abnormal

- Systemic exposure data and PD results for the vaginal ring formulation in this study demonstrated a trend for increasing endometrial transformation rates with increasing serum progesterone concentrations. However, serum concentrations were assessed using clinical chemistry methods and blood sampling times were not controlled. In addition, as noted earlier the correlation between a successful endometrial transformation (PD) and the primary clinical outcome (i.e. pregnancy) is not established.

Note: Serum progesterone and PD data available from two other phase 2 studies in this NDA (DR-PGN-201 & DR-PGN-202) are not being discussed in detail here. These studies were conducted using vaginal rings from a third manufacturing site (Barr Lab's Plainsboro, NJ facility) for which no bridging BE information is available and hence the clinical relevance of the findings is unknown. Results are however, summarized in Appendix at the end of this review.

2.2.4.2 Does this drug prolong the QT or QTc interval?

Thorough QT studies have not been conducted for the proposed vaginal ring formulation. The active ingredient is progesterone, an endogenous steroidal hormone.

2.2.5 What are the PK characteristics of the drug and its major metabolite?

2.2.5.1 What are the single dose and multiple dose PK parameters and how do PK parameters change with chronic dosing?

Study 10716222 evaluated the single dose and steady-state PK of the progesterone vaginal ring formulation (b) (4) in 30 healthy post-menopausal volunteers (27 completers). This was an open-label, one-treatment, two dose (2 x 7 day continuous cycles) study. Subjects received pre-treatment with 1 mg oral estradiol tablets once a day for 28 days prior to treatment and continued to receive estradiol treatment during the study period. Extensive sampling for PK was conducted at various time points after the placement of first and second rings and following removal of the second ring for capturing the drug elimination. Pre-dose samples were obtained at -48 h, -24 and 0 hours for baseline progesterone assessment. Baseline correction was done prior to analyses by subtracting the individual means of the three pre-dose samples from each of the post-dose concentrations.

Individual and average plasma progesterone (PGN) concentrations are shown following the two weekly doses:

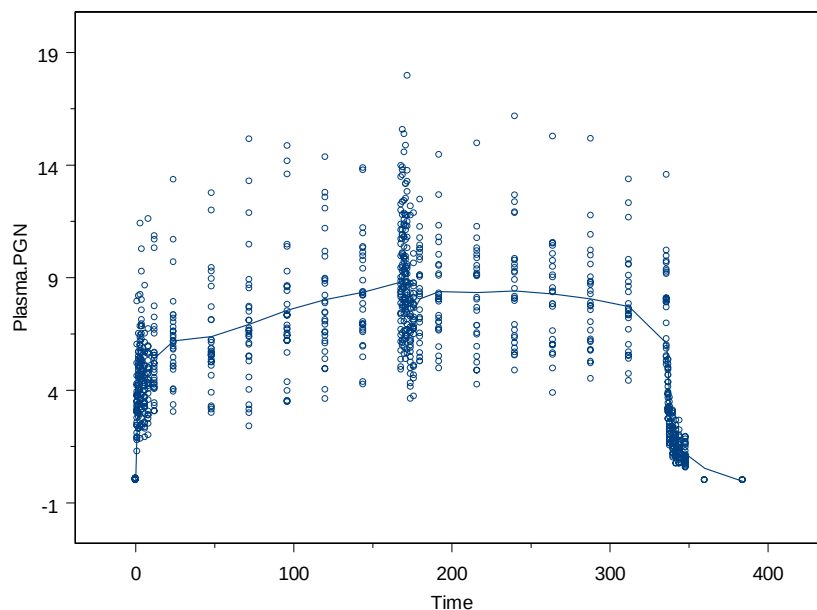


Figure 4: Scatter plot of plasma progesterone concentrations (ng/mL) vs. time (h) following two consecutive weekly vaginal ring doses in healthy post-menopausal volunteers of definitive PK study 10716222. The solid line represents mean data.

Table 6: Progesterone pharmacokinetics (baseline corrected) following the first and second vaginal rings

Pharmacokinetic Parameter	Units	N	Mean	SD	Median	Min.	Max.
First DR-2011 vaginal ring insertion							
C _{max} (0-168)	ng/ml	30	9.33	2.80	8.56	5.02	15.15
T _{max} (0-168)	hr	30	134.80	49.17	144.00	4.02	168.00
AUC ₀₋₁₆₈	ng·hr/ml	30	1188.41	374.25	1135.60	606.20	2142.33
Second DR-2011 vaginal ring insertion							
C _{max} (168-336)	ng/ml	27	10.66	2.72	10.36	6.63	17.96
T _{max} (168-336)	hr	27	206.15	56.33	172.00	168.00	336.00
AUC ₁₆₈₋₃₃₆	ng·hr/ml	27	1377.59	362.00	1401.27	827.69	2427.82
C _{max} (168-384)	ng/ml	27	10.66	2.72	10.36	6.63	17.96
T _{max} (168-384)	hr	27	206.15	56.33	172.00	168.00	336.00
AUC ₁₆₈₋₃₈₄	ng·hr/ml	27	1402.18	367.08	1431.24	843.41	2460.29
AUC _{168-inf}	ng·hr/ml	22	1382.62	387.81	1425.07	850.97	2476.21
K _e	hr ⁻¹	22	0.07	0.02	0.08	0.03	0.11
Elimhalf	hr	22	10.82	4.27	9.09	6.59	24.91

DR-2011: Progesterone vaginal ring

PK Conclusions:

- Baseline progesterone concentrations were low in the post-menopausal study population, with individual concentrations ranging from 0-0.15 ng/mL.
- Plasma progesterone concentrations steadily increased following the insertion of the first vaginal ring.
- Concentrations observed at the end of the first 7-day dosing interval were sustained by the insertion of the second ring at day 8. Thus, the plasma progesterone concentrations appear to have reached steady-state by the second dose of the progesterone vaginal ring at day 8.
- Based on AUC_{0-168h} values after the first dose (day 1) and the second dose (day 7), some accumulation of progesterone in plasma was evident, with a mean value of 1.26 fold. Accumulation in individual subjects was varied and ranged between no accumulation to accumulation of up to 3-fold.
- Peak progesterone values (C_{max}) after the first dose and second doses of the vaginal ring averaged at 9.3 ng/mL and 10.7 ng/mL, respectively. Median T_{max} values for the first and second rings were 144 hours and 172 hours, respectively.
- The average value of steady-state plasma progesterone (C_{avg}) observed during the second dose of the ring was 8.19 ng/mL.
- The inter-individual variability (% CV) in plasma progesterone C_{max} and AUC following the vaginal ring product was ~25- 30 %.

- Progesterone concentrations in plasma declined with a terminal T1/2 of ~ 11 hours following removal of the second ring at day 15.

Overall conclusion: The proposed vaginal ring formulation resulted in consistent release and absorption of progesterone via the vaginal mucosa into the systemic circulation.

2.2.5.2 How does the PK of the drug and its major active metabolites in healthy volunteers compare to that in patients?

PK data from detailed investigations in healthy post-menopausal volunteers are shown in the section above. In addition, similar data are available in this population from the bioequivalence trial 10936035 (see General Biopharmaceutics section of this review).

Serum progesterone concentrations were also measured using standard clinical chemistry methods in the phase 2 and phase 3 clinical trials in the target population (infertile premenopausal women undergoing ART). In women who become pregnant during the ART procedures, the endogenous progesterone concentrations continue to increase markedly thus confounding the contribution of exogenous progesterone.

Phase 3 clinical trial DR-PGN-302: Serum progesterone concentrations were evaluated at visits 4, 6, 9 corresponding to pre-dose, week 2 and week 10, after dose (first vaginal ring) initiation.

Serum progesterone concentrations (ng/mL) during these visits, further stratified by final pregnancy outcome are shown for this trial. Note that the time of blood draws (relative to dosing) for assessing serum progesterone concentrations were not controlled in the phase 3 trial. Patients self-administered the vaginal ring (one per week) or the Crinone vaginal gel (90 mg once daily in the morning) at home.

Table 7: Serum progesterone concentrations in pregnant and non-pregnant women of phase 3 study DR-PGN-302 at various visits up to the end of treatment (10 weeks).

	Serum progesterone concentrations (ng/mL)			
	(b) (4) Vaginal Ring once weekly		Crinone 8 % 90 mg qd	
	Non-pregnant (n = 301-334)	Pregnant (n = 294 -300)	Non-pregnant (n = 314-342)	Pregnant (277-289)
Visit 0 [screening]	1.38 ± 1.75	2.05 ± 4.55	1.73 ± 3.29	2.39 ± 2.86
Visit 4 [prior to first dose]	26.6 ± 17.1	27.85 ± 19.85	26.28 ± 16.04	29.11 ± 21.36
Visit 6 [at ~ 2 weeks post-dose]	10.76 ± 20.42	77.4 ± 83.26	9.76 ± 26.93	80.0 ± 79.7
Visit 9 [at the end of 10 weeks]	5.98 ± 27.74	57.61 ± 34.1	4.51 ± 18.56	61.15 ± 34

In the target population undergoing ART, serum progesterone concentrations were initially low. Concentrations in all women were subsequently higher than

baseline at visit 4(day after egg retrieval), even prior to the administration of investigational progesterone products. This could be the result of ovarian stimulation by the exogenously administered β HCG for final follicular maturation.

Concentrations of progesterone in women who subsequently became pregnant increased further, probably due to ovarian progesterone contribution from pregnancy HCG, while concentrations from non-pregnant women dropped probably due to diminishing contribution from the ovarian stimulation by exogenous HCG. By the end of the treatment period (10 weeks), serum progesterone concentrations in non-pregnant women continued to decrease, while those in pregnant women were still higher as is seen typically during pregnancy.

No trends were apparent between serum progesterone concentrations and the pregnancy outcomes in this study. In women who became pregnant during the ART procedures, the endogenous progesterone concentrations continued to increase markedly thus confounding the contribution of exogenous progesterone. The serum concentrations in women who failed to become pregnant appeared to be within the expected range previously noted for this new ring formulation and therefore do not readily provide an explanation to the lack of efficacy.

Despite the differences in study populations (post-menopausal, pre-menopausal volunteers or target population), matrices (serum vs. plasma), analytical methodology (LC-MS/MS vs. standard clinical chemistry) the systemic progesterone concentrations from the vaginal ring at steady-state conditions were in general comparable across studies in this NDA. Note that the Cavg values reported here for the PK study 10716222 and BE study 10936035 were based on baseline-corrected data, while it appears that for the phase 2/3 studies values reported are not corrected for baseline. Baseline concentrations reported for the two phase 1 studies were low ($\sim 0.1 - 0.2$ ng/mL) due to the use of post-menopausal women with endogenously low progesterone. In the phase 2/3 population, pre-treatment with estrogen prior to dosing generally resulted in suppression of endogenous progesterone to low concentrations relative to screening ($\sim 0.8 - 1.5$ ng/mL across studies).

Table 8: A comparison of systemic progesterone concentrations from the vaginal ring product across various studies of NDA 201110

Study	Vaginal Ring mean steady state concentration (ng/mL)	Comments
10716222	8.20	C _{avg,168-336} , plasma
10936035	6.75	C _{avg,96-168} , plasma
DR-201-102	7.89	Visit 5 (serum)
DR-PGN-201	6.58	Visit 5 (serum)
DR-PGN-202	6.00	Visit 3 (serum)
DR-PGN-302	7.77	Visit 6 (serum)

2.2.5.3 What are the absorption, distribution, metabolism and excretion (ADME) characteristics of the drug?

The absorption and elimination characteristics of progesterone from the vaginal ring formulation are addressed in the sections above. Pertinent information is summarized here again along with other ADME information as proposed in the draft labeling by the sponsor:

Absorption: Progesterone plasma concentrations (b) (4) increased following the (b) (4) administration of (b) (4) vaginal (b) (4) to healthy postmenopausal females once a week for two weeks. Steady state concentrations were attained (b) (4)

Distribution: Progesterone is approximately (b) (4) % bound to serum proteins, primarily to serum albumin and corticosteroid binding globulin.

Metabolism: Progesterone is metabolized primarily by the liver largely to pregnanediols and pregnanolones, which are conjugated to glucuronide and sulfate metabolites. (b) (4)

Excretion: Progesterone undergoes both biliary and renal excretion. Following an injection of labeled progesterone, 50%-60% of the excretion of metabolites occurs via the kidney; approximately 10% occurs via the bile and feces. Overall recovery of the labeled material accounts for 70% of an administered dose. Only (b) (4) of unchanged progesterone is excreted in the bile. (b) (4)

2.2.5.4 What is the inter- and intra-subject variability of PK parameters in volunteers and patients, and what are the major causes of variability?

In the definitive PK study 10716222, the inter-individual variability in plasma progesterone PK following the vaginal ring product was ~25- 30 %. In the BE study 10936035, the inter-subject variability in progesterone PK was 25 % and intra-subject variability was 13 %. In the post-menopausal population employed in these two studies, endogenous progesterone is low and unlikely to contribute significantly to variability. In addition, all women were pre-treated with estrogen so that any inherent differences in their vaginal epithelium would be diminished and therefore will not significantly affect the absorption of the drug from the ring.

2.3 Intrinsic Factors

2.3.1 What intrinsic factors influence exposure (PK) and/or response, and what is the impact of any differences in exposure on efficacy or safety responses?

- No specific-population studies were conducted in support of this NDA. The proposed indication is in infertile, pre-menopausal women who are undergoing ART treatments for achieving pregnancy. The phase 3 trial excluded patients with significant disease including organ dysfunction such as renal or hepatic impairment. A majority of phase 3 patients were under 35 years of age, while there was some information from 35- 42 years. The phase 1 studies imposed a BMI cut-off of 32, while the phase 3 study imposed a BMI cut-off of 38. Enrollment was not restricted in terms of race, however a majority of patients were Caucasians.
- Age: The effect of age on systemic PK was not consistent in this NDA. There was no correlation of age with systemic progesterone in the BE study 10936035. However, with increasing age there was a trend for increase in the progesterone plasma AUC values ($p < 0.05$) in the PK study 10716222. There was no correlation between age and BMI in the vaginal ring group.
- The efficacy outcomes (clinical pregnancy rates) are summarized below by {arbitrary} age categories. Keeping in mind that the sample size in the highest age categories is small and the phase 3 trial was not powered to identify differences in subgroups, it appears that the clinical pregnancy rates were lower as age increased. The phase 3 patients were predominantly under the age of 35 years.

Table 9: Efficacy outcomes for the vaginal ring presented by [arbitrary] age categories

Age category in phase 3	Clinical Pregnancy rate at both 8 and 12 weeks
19.8 -23.7 [n = 12]	50 %
23.7 – 27.6 [n = 73]	50.68 %
27.6 – 31.5 [n = 212]	52.35 %
31.5 – 35.3 [n = 261]	44.44 %
35.3 – 39.2 [n = 65]	38.46 %
39.2 – 43.1 [n = 17]	23.53 %

- **Body mass index (BMI):** The effect of BMI on systemic PK was not consistent in this NDA. There was no significant correlation between BMI and systemic progesterone concentrations (AUC and C_{max}) in the PK study 10716222 ($p > 0.05$). In the BE study 10936035 however, there was a significant trend ($p < 0.05$) showing decreasing C_{max} and AUC values of progesterone with increasing BMI (mean BMI: 27 kg/m²; range: 20- 32) and body weight (mean: 160 lbs; range 119-201). In this study (as well as in the definitive PK study) the enrollment based on BMI was restricted to a maximum of 32 kg/m².

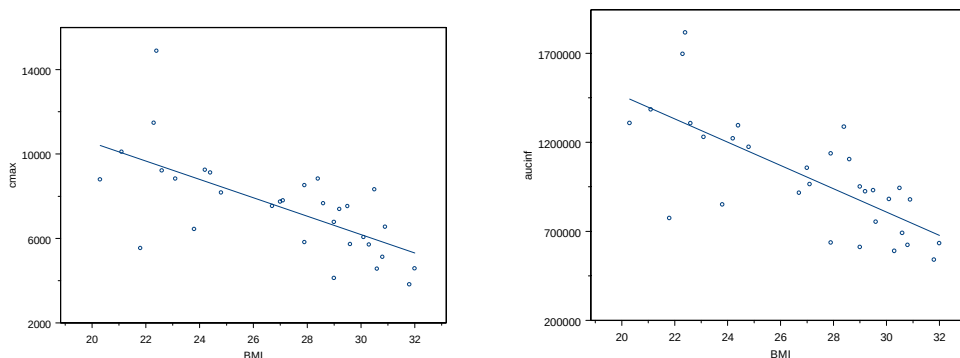


Figure 5: Trends for lower systemic PK of progesterone from the vaginal ring with higher BMI

Table 10: Presentation of systemic PK parameters by [arbitrary] BMI categories [study 10936035]

BMI category	AUC _{inf} ; ng h/mL	C _{max} ; ng/mL
20.18 – 22.17 [n = 3]	1155 ± 332	8.14 ± 2.35
22.17 – 24.16 [n = 5]	1379 ± 386	10.16 ± 3.18
24.16 - 26.15 [n = 3]	1229 ± 60	8.85 ± 0.59
26.15 – 28.13 [n = 5]	941 ± 190	7.48 ± 1.0
28.13 – 30.13 [n = 8]	929 ± 204	6.76 ± 1.44
30.13 – 32.12 [n = 7]	699 ± 152	5.52 ± 1.52

- **Discussion of potential clinical implication:** In the phase 3 clinical trial DR-PGN-302, patients were to be included if they had a BMI less than 38 kg/m². Analysis of pregnancy outcome data based on [arbitrary] BMI categories (shown in table below) suggests comparable pregnancy rates for BMI values up to ~34.5 kg/m². In the highest BMI category [34.5 – 38.5], the rates of pregnancy were lower with the progesterone vaginal ring formulation. While a similar trend was noted also for the vaginal gel formulation (active control), it was less pronounced than with the vaginal ring treatment.

Table 11: Presentation of pregnancy rates in phase 3 trial by [arbitrary] BMI categories

BMI ranges	Pregnancy rates % for vaginal ring (clinical pregnancy at both 8 and 12 wks)
15.5 – 19.5 [n = 38]	47.3
19.5 – 23.3 [n = 224]	48.6
23.3 – 27.0 [n = 175]	49.0

27.0 – 30.75 [n= 111]	45.9
30.75 – 34.5 [n = 55]	43.6
34.5 – 38.2 [n = 35]	28.5

- Sponsor's analysis of clinical pregnancy data shown below (not powered to show differences in subgroups) also suggests a similar trend for lower pregnancy rates (shown separately for the 8 weeks and 12 weeks) for the higher BMI group:

Table 12: Sponsor's presentation of efficacy data by BMI categories

	Weeks of Pregnancy	DR-2011 N=646 ^a		Crinone [®] N=651		DIFF (DR 2011- Crinone [®])	95% C.I. for DIFF
		N	%	N	%		
BMI <18.5 kg/m ²	8	11	5 (45.5%)	18	9 (50%)	-4.5%	(-42.0%, 32.9%)
	12		5 (45.5%)		9 (50%)	-4.5%	(-42.0%, 32.9%)
BMI ≥18.5 to <25 kg/m ²	8	343	171 (49.9%)	331	161 (48.6%)	1.2%	(-6.3%, 8.8%)
	12		165 (48.1%)		155 (46.8%)	1.3%	(-6.3%, 8.8%)
BMI ≥25 to <30 kg/m ²	8	186	90 (48.4%)	168	78 (46.4%)	2.0%	(-8.5%, 12.4%)
	12		89 (47.8%)		76 (45.2%)	2.6%	(-7.8%, 13.0%)
BMI ≥30 to <34 kg/m ²	8	65	29 (44.6%)	80	35 (43.8%)	0.9%	(-15.4%, 17.1%)
	12		27 (41.5%)		34 (42.5%)	-1.0%	(-17.1%, 15.2%)
BMI ≥34 to <38+ kg/m ²	8	39	14 (35.9%)	54	24 (44.4%)	-8.5%	(-28.6%, 11.5%)
	12		13 (33.3%)		20 (37.0%)	-3.7%	(-23.3%, 15.9%)

- In the phase 3 trial, there was also no correlation between age of the patient and their BMI value, thus suggesting absence of co-dependence of these two factors.

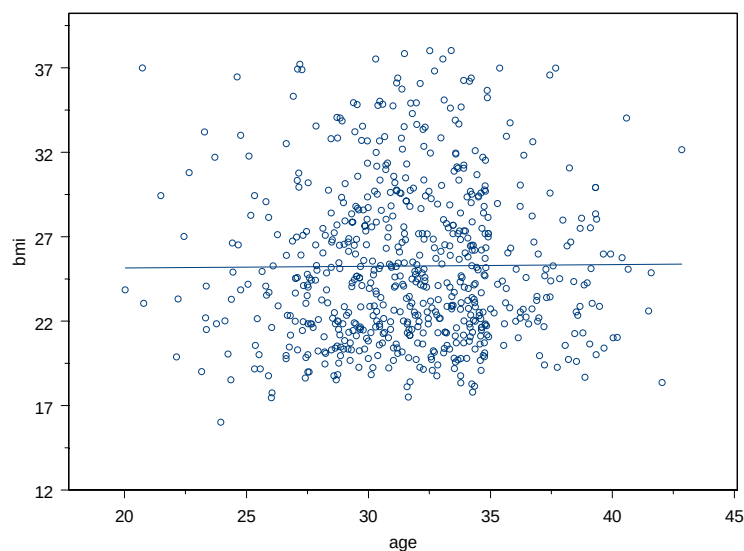


Figure 6: Absence of correlation between age and BMI in the phase 3 population.

- The limitations of these observations related to BMI are that the correlation between serum progesterone concentrations and clinical pregnancy is not well established and that the phase 3 trial was not powered to identify differences in subgroups. Reviewer nevertheless recommends that the proposed labeling for the vaginal ring formulation (if approved) should contain the following information regarding BMI in the ‘USE IN SPECIFIC POPULATIONS’ subsection:
- “The efficacy (b) (4) in patients with BMI above 38 kg/m2 has not been studied”.
- Race: In the definitive PK study 10716222 in healthy post-menopausal volunteers, there was no trend towards exposure differences with race.
- In the phase 3 clinical trial DR-PGN-302, there was no effect of race on the systemic exposure (visit 6 data shown below by race and pregnancy outcome); majority of the phase 3 population was comprised of Caucasians:

Table 13: Summary of serum progesterone by pregnancy status and by race in phase 3 trial

Race	Non pregnant (ng/mL)	Pregnant (ng/mL)
Asian	7.2 ± 1.55	64.35 ± 69.85
African-American	11.1 ± 9.43	63.12 ± 63.39
Hispanic	8.8 ± 6.92	43 ± 66
Caucasian	16.27 ± 33	72 ± 82

- The breakdown of clinical response (clinical pregnancy rates) by race in the phase 3 trial is shown below; majority of the population is Caucasian. A somewhat lower pregnancy rates were noted in the African-American population, although the study is not powered to identify these differences in subgroups.

Table 14: Summary of efficacy data by active treatment and by race in phase 3 trial

	Clinical Pregnancy at both 8 and 12 weeks	
	Vaginal Ring (b) (4)	Crinone 8 % Gel
Asian	44.1 % [n = 34]	37.8 % [n = 37]
African-American	28.3 % [n = 53]	47.8 % [n = 47]
Caucasian	47.8 % [n = 514]	46.2 % [n = 502]
Hispanic	61.7 % [n = 34]	48.0 % [n = 50]

Conclusions: The labeling should contain the following information if the drug is approved:

- Efficacy and safety have not been studied in patients with BMI >38 kg/m2.
- Pharmacokinetics in presence of renal and hepatic impairment have not been assessed.

2.4 Extrinsic Factors

2.4.1 What extrinsic factors influence dose-exposure and/or -response and what is the impact of any differences in exposure on response? Based upon what is known about exposure-response relationships and their variability, what dosage regimen adjustments, if any, do you recommend for each of these factors?

No specific drug-drug interaction studies have been conducted with the progesterone vaginal ring formulation. With the exception of concomitant vaginal formulations such as prescription or over-the-counter creams, gels or suppositories, there were no other specific exclusions with regard to concomitant drugs. Subjects were also not to have used tobacco within the past three months.

Listing of concomitant medications in phase 3 included antihistaminics, prenatal vitamins, pain medications, antacids, antibiotics, cough suppressants, topical local anesthetics, anti-nausea medications, thyroid medications etc. Many of the co-medications were part of standard of care during ART procedures.

While metabolism of progesterone is thought to occur primarily via reduction processes, there is literature evidence to indicate oxidative metabolism by CYP3A4, CYP2C19 and to some extent CYP2C9 (Swinney et al, 1990; Yamazaki et al, 1997). There are no in vivo DDI studies to document that enzyme inhibitors can cause increased plasma levels of progesterone. In vitro data by Swinney et al showed that the metabolism of progesterone in human liver microsomes was inhibited by Ketoconazole. Due to the potential involvement of CYP450s, there is at least a theoretical likelihood that enzyme inhibitors may alter the metabolism of progesterone. However progesterone formulations provided much higher systemic exposure than the proposed vaginal drug product have been approved (e.g. intramuscular injection).

The likelihood of an enzyme inducer increasing the clearance of progesterone from the vaginal ring is also theoretical. There are no reports in the literature that suggest that enzyme inducers can alter therapeutic progesterone levels achieved in ART by IM, oral or vaginal administration.

The likelihood of a clinically relevant effect of progesterone on the metabolism of concomitant medications appears to be low as literature survey did not yield in vivo evidence of such interactions. There is some information to suggest that in vitro, at concentrations much higher than seen for vaginal ring product, progesterone inhibits CYP2C19 and CYP2C9. The clinical relevance of such data is not known.

Concomitant use of other vaginal products: Concomitant use of other vaginal formulations such as anti-fungal creams may in theory have an effect on the drug release and absorption from the vaginal ring product. In the phase 3 clinical trial for the proposed formulation, concomitant use of vaginal products was not allowed. Due to the limited duration of use for this indication (10 weeks if pregnancy is established), it is possible that by avoiding the concomitant vaginal

medications, any potential implications on the drug release from the vaginal ring may be avoided.

Conclusions: The following language has been proposed by the sponsor in the draft labeling with respect to drug interactions and concomitant vaginal product usage is found acceptable. The language proposed is similar to that approved in the package insert for another progesterone vaginal product Endometrin (progesterone vaginal inserts):

DRUG INTERACTIONS

No formal drug-drug interaction studies have been conducted for (b) (4). Drugs known to induce the hepatic CYP3A4 system (such as rifampin, carbamazepine) may increase the elimination of progesterone. The effect of concomitant vaginal products on the exposure of progesterone from (b) (4) has not been assessed. (b) (4) is not recommended for use with other vaginal products (such as antifungal products) as this may alter progesterone release and absorption from the vaginal (b) (4).

2.5 General Biopharmaceutics

2.5.1 Is the proposed to-be-marketed formulation bioequivalent to the pivotal clinical trial formulation of progesterone vaginal ring?

Yes, the proposed commercial formulation, manufactured at Barr Lab's Cincinnati, OH commercial facility was demonstrated to be bioequivalent to the clinical trial formulation, manufactured at Barr Lab's Northvale, NJ R & D facility) as seen from Study 10936035 conducted in healthy post-menopausal females. The two products differed only in the site of manufacturing, while there were no changes to the formulation itself. However, during pre-NDA discussions with the sponsor, the Office of New Drug Quality Assessment (ONDQA) recommended that the sponsor should conduct a formal BE study to bridge the two manufacturing sites.

The BE study design was that of an open-label, single dose (7-day vaginal ring), multiple site, two-treatment, two-period, two-sequence, randomized crossover study in N= 32 post-menopausal women.

Subjects received a single dose of once-weekly vaginal ring formulation- either Reference (clinical trial formulation; manufactured at the R & D facility in NJ; batch number 900214) or Test (commercial formulation; manufactured at commercial facility in OH; batch number 800357) in period I or period II of the study separated by a 7-day washout duration. All subjects received pre-treatment with 1 mg estradiol oral tablets for at least 28 days prior to first dose and also continued with this treatment throughout the entire study including the washout period, until after the final blood draw in period II.

PK sampling and analysis: Blood samples for plasma progesterone analysis were collected at pre-dose (for baseline; -1.0, -0.5 and 0 h), extensively during the two treatment periods and after removal of the vaginal ring in the second period.

Samples were analyzed by (b) (4) using a validated LC-MS/MS methodology. All post-dose samples were corrected for baseline by deducting the mean of three pre-dose values (-1, -0.5 and 0 hours). PK analyses were conducted on the baseline-corrected data. Non-baseline corrected data analysis is presented as supportive data. Bioequivalence was based on the 90 % confidence intervals surrounding treatment mean ratios (T: R) for AUC0-t, AUC0-inf and Cmax.

Results: Box and whisker plots of progesterone PK [Cmax: pg/mL; AUCinf: pg.h/mL] for the test (A) and reference (B) formulations are shown:

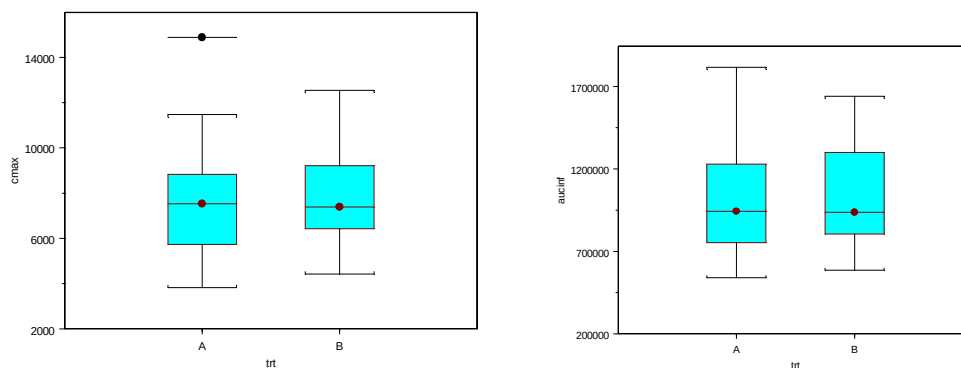


Figure 7: Box and whisker plots of Cmax and AUC for the test (A) and reference (B) formulation in study 10936035

Figure 8: Overlaying profiles of the progesterone concentration vs. time in BE study 10936035.

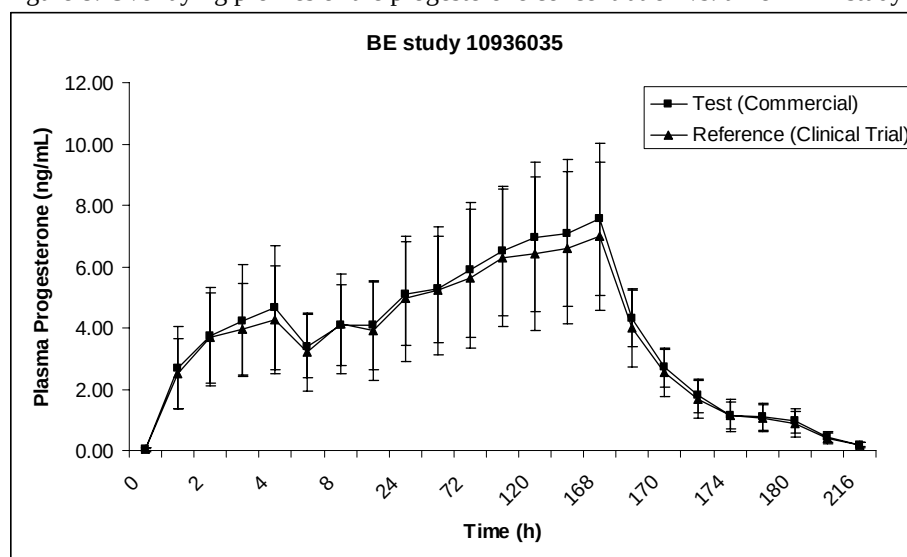


Table 15 (repeat of Table 1): BE analyses outcomes in study 10936035 (WinNonlin 5.2)

Test = A (Proposed commercial); Reference = B (Clinical trial formulation)

	Test GeoLSM	Ref GeoLSM	Ratio [% Ref]	90 % CI range
C _{max} (ng/mL)	7.14	7.54	94.62	89.37 - 100.18
AUC (ng h/mL)	949.35	1002.21	94.73	89.51 - 100.25
AUC _{inf} (ng h/mL)	952.53	1005.53	94.73	89.52 - 100.24

The point estimates (expressed as % reference B in the table above) for all three PK parameters above were ~ 95 %, while the 90 % confidence intervals surrounding these ratios were contained entirely within the bioequivalence limits of 80-125 %. Thus the two treatments meet the bioequivalence criteria.

Bioequivalence analysis outcomes were also similar for the uncorrected progesterone data, with the point estimate for Test/Reference ratio of PK parameters C_{max}, AUC and AUC_{inf} close to 100 % and their associated 90 % CI ranges within 80-125 % bioequivalence bounds.

A summary of the PK parameters is given below. The inter-subject variability was ~ 25 % and intra-subject variability was ~ 13 % in this study.

Table 16: A summary of progesterone PK in the BE study 10936035

PK Parameter	Test A (Commercial batch)	Reference B (Clinical trial batch)
AUC _{0-t} (ng.h/mL)	1005 ± 315 (31 %)	1045 ± 283 (27 %)
AUC _{0-inf} (ng h/mL)	1009 ± 316 (31 %)	1049 ± 283 (27 %)
C _{max} (ng/mL)	7.5 ± 2.3 (30 %)	7.8 ± 2.0 (26 %)
T _{max} (h)	146 ± 33 (12 %)	143 ± 37 (23 %)
T _{1/2} (h)	14.5 ± 1.7 (12 %)	14.8 ± 2.4 (15.7 %)

Bioequivalence analysis suggested significant period effects in this study ($p < 0.05$). Data suggested that systemic exposure for both treatments (test A or reference B) was higher during period 1 compared to that in period 2, suggesting a potential period effect on both treatments.

Certain aspects of the BE study design were adequately planned to account for any potential issues such as these. Firstly, all study subjects (post-menopausal women) were dosed with oral estrogen for 28 days prior to period 1 (and continued throughout the study) to normalize any differences in vaginal atrophy

which is more prevalent in this population. In addition, the sponsor also allowed 7-day wash-out duration between the two study periods, thereby avoiding concerns related to drug-carryover and potentially pharmacological effect carryover into the second period. In addition, since the current study was a 2-sequence crossover study, half of the subjects received treatment A (test) followed by treatment B (reference) while the other half received B followed by A. In essence any change that might favor B over A in one sequence will favor A over B in the other sequence and cancel out of the treatment comparison.

To address the potential period effect further, a separate BE analysis was conducted only for the data collected in period 1; thus the resulting comparison will be that of a parallel group comparison of treatment A and treatment B since the alternate treatment received by each of the subjects in period 2 of the study has been excluded. Data suggests bioequivalence of the two treatments when administered in parallel groups to a total of 32 subjects.

Table 17: BE analysis of period 1 data only from study 10936035

Period 1 only	Test GeoLSM	Reference GeoLSM	Ratio [% Ref]	90 % CI range
Cmax (ng/mL)	7.79	7.69	101.26	85.66 - 119.71
AUC (ng h/mL)	1059.48	1067.75	99.23	83.58 - 117.80
AUCinf (ng h/mL)	1062.21	1071.33	99.15	83.53 - 117.68

One potential explanation for a period effect in this case that was considered during this review was whether following a 7-day in vivo residence of the vaginal progesterone ring during the first period and continuous exposure to progesterone locally, there might have been some local changes (thickening) to the vaginal mucosa resulting in lower drug absorption from this site during period 2 of the study.

However, in a definitive PK study 10716222 involving two consecutive doses of the test drug formulation (using same batch as in the BE study; similar post-menopausal population and similar pre-treatment with estradiol), there was no evidence of decreased drug absorption during a second dose of the vaginal ring formulation that was administered immediately following a 7-day in vivo residence of the first vaginal ring dose. In fact a modest accumulation of the drug (1.26-fold average) was seen in this study following the second dose [Cmax values of 9.3 ng/mL and 10.7 ng/mL, respectively]. Thus earlier doses do not seem to have an effect on the drug absorption from subsequent doses.

Overall, it does not appear that the period effects observed would have any meaningful influence on the BE outcomes of this study. Desirable features of the study design such as 2-way cross-over of treatments, use of an adequate wash-out duration between test and reference, and use of pre-treatment with estradiol,

coupled with an acceptable BE outcome when only period 1 data were analyzed further, suggest acceptability of the overall information presented. Hence the study is acceptable in establishing bioequivalence of the clinical vs. to-be-marketed formulations of the vaginal ring.

2.5.2 What is the apparent in vivo release rate of progesterone from the new vaginal ring formulation?

During the BE study 10936035 and a follow-up ring exposure study 10936057, progesterone vaginal rings were collected after one week use period from a total of 45 subjects for residual drug content analysis. A control group of 45 rings from the same batch were used to derive the initial amount of drug in the vaginal rings.

The difference between the least squares means of the unused and used rings was 75.44 mg released over 7 days and the 95% confidence interval around this difference was 68.84 – 82.05 mg. Expressing these results as average release on a daily basis provides a release rate of 10.78 mg per day with a 95% confidence interval of 9.83 – 11.72 mg/day.

Thus the sponsor proposes to include an average release rate of 11 mg/day in the labeling.

While prior analysis in this regard by the sponsor yielded varying results for in vivo drug release, ranging from 6 – 15 mg/day, due to the design aspects of this most recent study, the results appear to be more reliable. For example, the sample size is higher than any other assessment done before (45 test rings from 45 subjects and 45 unused control rings), and the rings were all part of the BE clinical trial and its follow-up study thus allowing dosing and retrieval of rings in a more controlled setting. In addition, the sample analysis was done in a way to avoid inter-day or inter-occasion variability in analysis. Therefore, it is reasonable to employ residual drug content analysis results from this study to compute the estimated in vivo release rate of the progesterone vaginal ring.

Sponsor has furthermore estimated the average concentrations (C_{avg}) of progesterone over each of the seven 24 hour intervals (by dividing the 24 hour AUC data for the second ring by 24) in the PK study 10716222:

Table 18: Calculated average daily in vivo plasma progesterone concentrations in study 10716222

Parameter	Progesterone [Mean $\pm$ SD (% CV)] (ng/mL)
$C_{avg168-192}$ (day 1)	8.23 $\pm$ 1.95 (23.7 %)
$C_{avg192-216}$ (day 2)	8.18 $\pm$ 2.18 (26.6 %)
$C_{avg216-240}$ (day 3)	8.26 $\pm$ 2.46 (29.7 %)
$C_{avg240-264}$ (day 4)	8.27 $\pm$ 2.42 (29.3 %)
$C_{avg264-288}$ (day 5)	8.06 $\pm$ 2.31 (28.7 %)
$C_{avg288-312}$ (day 6)	7.84 $\pm$ 2.15 (27.5 %)

$C_{avg312-336}$ (day 7)	7.73 ± 1.96 (25.4 %)
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Based on these results sponsor notes that the in vivo drug delivery rate (and therefore the in vivo release rate) must have been steady throughout the entire 7 day use period at steady-state, thus justifying estimation of in vivo release rate by dividing the difference between the initial and residual drug content by 7 days (the in vivo residence time of these rings).

Overall data suggest that the ring releases approximately 75 mg (4.2 % of its 1.78 g drug load) over the 7 days of in vivo residence time. This translates to ~ 11 mg/day of in vivo drug release rate resulting in an average concentration (C_{avg}) of ~ 8 ng/mL per day at steady-state.

There is no established IVIVC for this drug product. The proposed in vitro release test is for quality control purposes and has no in vivo relevance.

2.6 Analytical Section

PK Study 10716222: Quantitation of progesterone in human plasma from this study was conducted using a validated HPLC method with MS/MS detection. The method validation and sample analyses were conducted by (b) (4)

Validation: The method is applicable to the quantitation of progesterone within a nominal range of 0.05 to 50.0 ng/mL. Correlation coefficient (average) of three calibration curves was > 0.990. Accuracy and precision data for back-calculated calibrators was also within acceptable limits.

Precision and accuracy of the assay were evaluated during validation by analyzing surrogate matrix QC pools prepared at 0.05, 0.15, 0.4, 1.5, 6.0, and 37.5 ng/mL. Additional QCs were prepared in human plasma at baseline, low and high levels (QC10, QC 7 and QC8). Precision was expressed as the percent coefficient of variation (%CV) of each pool. Accuracy was measured as the percent difference from theoretical. Results for precision and accuracy were found to be within pre-determined acceptance criteria set forth by assay validation protocols and SOPs.

Table 19: Accuracy and precision summary for QCs during the validation experiment for study 10716222

	QC 0 0.05 ng/mL	QC 1 0.15 ng/mL	QC 2 0.4 ng/mL	QC 3 1.5 ng/mL	QC 4 6.0 ng/mL	QC 5 37.5 ng/mL	QC 7 0.218 ng/mL	QC 8 37.6 ng/mL	QC10 Baseline
Mean	0.0535	0.162	0.419	1.56	6.26	37.9	0.228	40.2	0.0676
Precision % CV	12.5	6.23	6.11	3.56	4.14	2.99	5.84	3.12	11.0
Accuracy % diff from theoretical	6.96	8.11	4.79	4.14	4.26	0.994	4.46	6.99	N/A

Dilution parallelism: Ability to analyze samples with insufficient volume or samples above the upper limit of calibration range after diluting was analyzed with six replicates of 1.5 ng/ml surrogate matrix, 37.6 ng/ml of human plasma (5-fold dilution) and 250 ng/ml surrogate matrix (10-fold dilution). Data (accuracy and precision) of diluted samples was acceptable in relation to the theoretical concentrations of these standards.

Extraction recovery: Recovery was determined by comparing pre- and post-extraction spiked samples (6 replicates) during one of the validation runs. The % recovery at 0.15, 1.5 and 37.5 ng/ml concentrations was 66.6 %, 60 % and 70.3 %.

Stability: Stability after three freeze/thaw (F/T) cycles was assessed for surrogate matrix and human plasma QCs. Precision and accuracy data of these QCs was within predetermined acceptance criteria suggesting analyte stability after 3 F/T cycles. In addition, there were no issues with cross-analyte interference and carryover effects.

Thawed matrix stability, post-preparative extract stability, and frozen sample stability were all found to be acceptable. Frozen samples were stable for 148 days at -70°C or colder. All samples in this study were analyzed within the established stability timeframe.

Sample analysis: Samples were analyzed using the above validated method. There were no unresolved issues with the sample analysis. Precision and accuracy data for 48 QCs at five different concentrations included the various assay runs was found acceptable.

BE Study 10936035: Sample analyses were conducted by [REDACTED] (b) (4) [REDACTED] using a validated LC-MS/MS assay for the determination of progesterone in human plasma.

Validation results: Method is applicable for assay of progesterone concentrations ranging 0.05 to 30 ng/mL. > 0.997 correlation was achieved for the standard curves. Inter-assay precision and accuracy for calibration standards for progesterone were acceptable. Precision and accuracy for the quality control samples was acceptable. The accuracy (% difference from theoretical) for QCs [at 0.05(LLOQ), 0.15, 4.5, 22.5 and 30 (ULOQ) ng/mL] included in the five runs ranged from -9.4 % to 1.8 % and had a % CV (precision) of not more than 10.7 % for progesterone. Intra-assay precision and accuracy data for QC samples were also acceptable [% CV no more than 8 %; and % nominal within ± 4 %].

Recovery of progesterone was reproducible at each of the QCs evaluated. Dilution integrity evaluated based on six different dilutions at 5-fold and 10-fold dilutions from a theoretical 60 ng/mL concentration was acceptable. In addition, freeze thaw stability was acceptable after five cycles at -22°C and -80°C, and stability was ensured for at least 96 days when stored at -22°C. In process stability (for at least 2 h), processed sample stability (for at least 82 hours at RT after processing), autosampler (re-injection) stability of extracted samples (for up to 75 hours later) were acceptable.

Study sample analysis: From the date of the first sample collection until the last day of analysis, the clinical samples were stored at a temperature of $-22\pm 10^{\circ}\text{C}$ for a maximum of 57 days. The stability of Progesterone in human plasma for at least 96 days at $-22\pm 10^{\circ}\text{C}$ has been established. No stabilities were surpassed.

Quality control samples were analyzed in replicates of two at four concentration levels. Overall, the inter-assay precision (%CV) and accuracy (%Difference) ranged from 2.3% to 8.9% and from -1.3% to 0.7% , respectively.

In this study, 128 samples were analyzed as part of the incurred sample re-analysis evaluation. Two samples/subject/period (one at $C_{\text{max}}/T_{\text{max}}$ timepoint and one at elimination phase time point, for each period) were selected and analyzed. For Progesterone, 126 of the 128 samples (98.4%) were within $\pm 20.0\%$ of the mean of both re-assays, allowing us to conclude with confidence that the method was performed as validated and is reproducible. Overall study findings were acceptable.

Tissue progesterone determination: Quantitation of progesterone in human endometrium tissue in study DR-201-102 was achieved via a validated HPLC with MS/MS Detection [PPD]. The validation was conducted using samples prepared in two matrices: human endometrium tissue and a surrogate matrix containing bovine serum albumin (BSA) in phosphate buffered saline (PBS) that was also subjected to preparing and processing steps in an identical manner with the tissue matrix samples to eliminate bias. The final method was applicable for the quantitation of progesterone from a nominal 50.0-mg human endometrium tissue aliquot within the nominal range of 0.100 to 50.0 ng/g. In addition to progesterone analysis, the total protein content of each tissue matrix sample was measured using the standard Pierce Bicinchoninic Acid (BCATM)-based protein assay.

Progesterone results for tissue samples were initially obtained on a wet-weight basis by analysis of tissue digest using the validated LC-MS/MS assay. The total protein content was also measured for the tissue samples using a small aliquot from the same sample digest. The total protein concentrations were then divided into the wet-weight progesterone concentrations to obtain tissue progesterone values on a total protein content basis.

Validation results: Linearity and calibration: Eight surrogate matrix calibration standards were analyzed in duplicate over the nominal concentration range of 0.100 to 50.0 ng/mL (equivalent to 0.100 to 50.0 ng/g for tissue matrix). The average correlation coefficient from two standard curves was > 0.990 .

Limit of quantitation: For this validation, the lower limit of quantitation for progesterone was nominally 0.100 ng/mL in surrogate matrix, which is equivalent to 0.100 ng/g in tissue matrix.

Precision and accuracy: Precision and accuracy were evaluated by analyzing surrogate matrix quality control pools prepared at 0.100, 0.250, 0.600, 2.40, 8.00, and 38.0 ng/mL (QC 0, 1, 2, 3, 4, and 5, respectively). Additional tissue quality controls were prepared with low and high analyte levels by fortifying with an additional 0.250 and 38.0 ng/g of analyte (designated as QC 11 and 15, respectively).

Table 20: Inter-assay precision and accuracy in surrogate matrix for tissue progesterone assay

Run ID	QC 0 (ng/mL)	QC 1 (ng/mL)	QC 2 (ng/mL)	QC 3 (ng/mL)	QC 4 (ng/mL)	QC 5 (ng/mL)
1QIR5-B	0.106	0.252	0.649	2.44	8.31	39.7
	0.0978	0.288	0.606	2.47	8.26	38.7
	0.106	0.245	0.602	2.44	8.12	37.9
	0.0932	0.248	0.600	2.45	8.22	39.3
	0.184 X	0.218	0.646	2.55	8.23	39.0
	0.0850	0.258	0.631	2.41	8.17	37.8
2QIR5-A	0.0977	0.268	0.599	2.18	8.33	40.8
	0.101	0.264	0.636	2.53	8.39	40.0
	0.107	0.263	0.617	2.53	8.45	38.4
	0.112	0.276	0.627	2.54	8.57	40.8
	0.106	0.259	0.652	2.52	8.66	40.8
	0.104	0.297	0.625	2.57	8.39	40.5
N	11	12	12	12	12	12
Theoretical Concentration	0.100	0.250	0.600	2.40	8.00	38.0
Mean	0.101	0.261	0.624	2.47	8.34	39.5
S.D.	0.00763	0.0206	0.0195	0.106	0.161	1.12
%C.V.	7.53	7.88	3.13	4.30	1.93	2.83
% Difference from Theoretical	1.43	4.52	4.03	2.90	4.27	3.86
Low Limit	0.0800	0.213	0.510	2.04	6.80	32.3
High Limit	0.120	0.288	0.690	2.76	9.20	43.7

LEGEND:

X Value excluded is an outlier as determined by the Dixon (rank) test.

Parallelism precision and accuracy: The ability to dilute human endometrium tissue samples originally above the upper limit of the calibration range was validated by analyzing six replicate 250 ng/g QCs as ten-fold dilutions in Run 1QIR5-B.

Long-term analyte stability in frozen matrix was established for 191 days at -70°C. Analyte stability in thawed matrix also was adequately demonstrated when stored for 24 hours prior to analysis at room temperature.

3 Detailed Labeling Recommendations

The following changes to the proposed labeling in Physician's Labeling Rule (PLR) format are recommended from a Clinical Pharmacology perspective. Additions are shown in bold underlined font and deletions in strikethrough format:

HIGHLIGHTS OF PRESCRIBING INFORMATION

DOSAGE AND ADMINISTRATION

The dose of [TRADENAME] is one vaginal ring inserted (b) (4) the day after oocyte retrieval (b) (4) replaced weekly, continuing for up to 10 weeks total duration. (b) (4)
[2.1].

DOSAGE FORMS AND STRENGTHS

(b) (4) vaginal (b) (4) **1.78 g** progesterone (b) (4) ~~an average of 11 mg/day progesterone over (b) (4) 7 day period (b) (4) [3]~~

FULL PRESCRIBING INFORMATION

3. DOSAGE FORMS AND STRENGTHS

[TRADE NAME] ^{(b) (4)} a white to off-white, flexible, non-biodegradable, ^{(b) (4)} silicone ^{(b) (4)} ring ^{(b) (4)} an average of 11 mg/day ^{(b) (4)}

8. USE IN SPECIFIC POPULATIONS

8.6. Body mass index (BMI)

Efficacy of [TRADE NAME] in women with a BMI > 38 kg/m² has not been studied.

11. DESCRIPTION

When placed in the vagina, each ^{(b) (4)} is estimated to provide an average release rate of 11 mg/day of progesterone ^{(b) (4)} over 7 days ^{(b) (4)}.

12.3. Pharmacokinetics

^{(b) (4)}

4 Appendices

4.1 Individual Study Reviews

Phase 2 data employing earlier (non-phase 3) batches of the vaginal ring formulation: Serum progesterone concentrations and PD data from two phase 2 studies employing non-phase 3 batches of the vaginal ring (manufactured at ^{(b) (4)} site) are presented here. No bridging BE information is available to link this batch to the phase 3 or the proposed commercial batches (differing only in the site of manufacturing).

- Study DR-PGN-201: In this active-controlled, phase 2 study in women with clinical or medically induced agonadism (n = 10 per group), serum progesterone concentrations obtained by clinical chemistry methods

(sampling times uncontrolled) were higher in patients on the twice daily (BID) Crinone regimen, compared to the once-weekly vaginal ring.

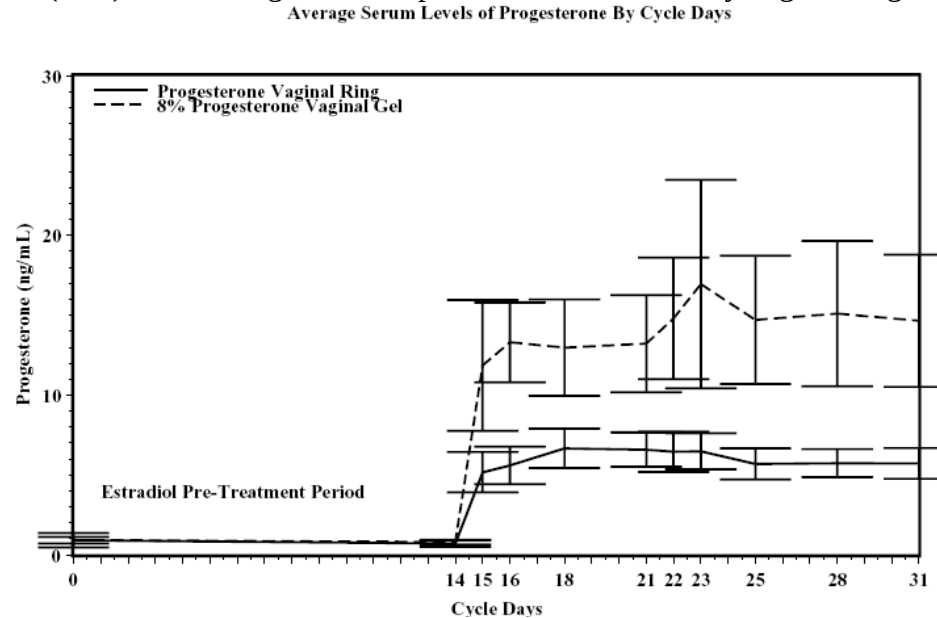


Figure 9: Serum progesterone across various visits in study DR-PGN-201.

- Note that in the other phase 2/3 trials of this NDA, the active control Crinone vaginal gel was used at a dose of 90 mg once daily, while it was used at a 90 mg twice daily dose in this phase 2 study. This is due to a difference in the populations evaluated i.e. those needing progesterone supplementation vs. those needing replacement (agonadal women). Crinone 8 % vaginal gel is approved for use both in women needing progesterone supplementation (at 90 mg once daily) or in women needing replacement due to partial or complete ovarian failure (at 90 mg twice daily) as part of ART treatment. The progesterone vaginal ring was evaluated only for the supplementation indication and not for replacement.
- The % patients achieving endometrial transformation was higher for the Crinone vaginal gel (100 %) compared to the ring (80 %). Two (out of ten) patients in the vaginal ring group who experienced an inadequate transformation of the endometrium had serum concentrations lower than 4 ng/mL.

Table 21: endometrial transformation rates in study DR-PGN-201

Endometrial Biopsy	Progesterone VR (N=10)		8% Progesterone Vaginal Gel (N=10)		Total
	N	%	N	%	
Adequate Secretory Transformation	8	(80.00)	10	(100.0)	18
Inadequate Secretory Transformation*	2	(20.00)	0	(0.00)	2

- Study DR-PGN-202: This phase 2 study was an extension of study DR-PGN-201 to evaluate clinical pregnancy rates in those patients who experienced successful endometrial transformation in the previous study.

Study was active-controlled in a total of 9 patients that had consented to participate following a successful conclusion of study DR-PGN-201. The clinical pregnancy rate determined by ultrasound at 8 weeks was 80 % for the vaginal ring treatment (4 out of 5 who underwent embryo transfer) and 25 % for the Crinone treatment arm (1 out of 4 who underwent embryo transfer). No definitive trends could be discerned as to the 'desirable' threshold of serum progesterone concentrations with respect to positive pregnancy outcomes. Also, evidence of a successful endometrial transformation (PD) in the earlier mock cycle (Study DR-PGN-201) using the same treatment regimen in the same patients did not ensure clinical success (i.e. pregnancy) in subsequent study 202 on the same regimen. However, the sample size is very small (n= 5 in vaginal ring group and n = 4 in the vaginal gel group) to make any definitive conclusions in this regard.

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/s/

SANDHYA K APPARAJU
02/02/2011

MYONG JIN KIM
02/02/2011